Linear Algebra 2

KNU Math 232

Classnotes

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These notes are for the second semester of a second year class in Linear Algebra based on the International Student's fourth edition of Gilbert Strang's 'Linear Algebra and its applications'. This is referred to as The Text. You SHOULD buy the text. It is full of examples and explanations that we will not have time for; and, I am assigning questions out of the text.

If you have the 4th edition of the text, but not the International Students 4th edition, then the problem numbers will not be the same. You should make sure you are doing the correct problems.

5 Eigenvectors and eigenvalues

5.1 Introduction

An *eigenvalue* of a matrix M is a constant λ such that

$$Mx = \lambda x$$

for some non-zero vector x. The vector x is the corresponding *eigenvector*.

Example 5.1. For the matrix $M = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]$, we have

$$M\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] \text{ and } M\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 \\ 3 \end{smallmatrix}\right] = 3\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right],$$

so 1 is an eigenvalue with eigenvector $(1, 0)$ and 3 is another eigenvalue with eigenvector $(0, 1)$. There are no more eigenvalues in this example. We will see soon that **an $n \times n$ matrix can have at most n eigenvalues.**

Practice

Does $M = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]$ does have more **eigenvectors**?

Indeed $M\left[\begin{smallmatrix} 7 \\ 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 7 \\ 0 \end{smallmatrix}\right]$. All vectors in the space $\langle(1, 0)\rangle = \{r(1, 0) \mid r \in R\}$ are eigenvectors with eigenvalue 1.

Practice

Show that the set of vectors that are eigenvectors for a given eigenvalue λ are a space.

This space is the *eigenspace* of λ.

So $M = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]$ has eigenvectors in the spaces $\langle(1, 0)\rangle$ and $\langle(0, 1)\rangle$.

Practice

Show that M has no other eigenvectors.

Practice

What are the eigenvectors of the rotation $M = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$?

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In this example it was easy to find the eigenvectors by inspection. This is not always the case. What are the eigens (eigenvalues and eigenvectors) of $M = \left[\begin{smallmatrix} 2 & 3 \\ 1 & 4 \end{smallmatrix}\right]$? Well we want v such that $Mv = \lambda v$ for some λ. We can write this as the matrix equation $Mv = \lambda I v$, and re-arrange this to get

$$(M - \lambda I)v = 0.$$

So we are looking for λ such that the *characteristic matrix* $M - \lambda I$ of M is singular. Its nullspace is the eigenspace of λ. As a matrix is singular if and only if its determinant is 0, we can find the eigenvalue λ for M as follows.

Example 5.2. To find the eigens of $M = \left[\begin{smallmatrix} 2 & 3 \\ 1 & 4 \end{smallmatrix}\right]$ we write out its *characteristic equation*:

$$0 = |M - \lambda I| = \begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix}.$$

Using our determinant formula to expand on the right to find the *characteristic polynomial* of M:

$$\begin{aligned} \begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix} &= (2-\lambda)(4-\lambda) - 3 \\ &= \lambda^2 - 6\lambda + 5 \\ &= (\lambda - 1)(\lambda - 5) \end{aligned}$$

solve the quadratic equation $(\lambda - 1)(\lambda - 5) = 0$ to get $\lambda = 1$ or 5.

We usually write the eigenvalues in increasing order as $\lambda_1 = 1$ and $\lambda_2 = 5$.

Practice

Find the eigenspaces of $\lambda_1 = 1$ and $\lambda_2 = 5$ by finding the nullspaces of $M - \lambda_1 I$ and $M - \lambda_2 I$.

The same process works for any square matrix M:

- Find eigenvalues λ_i by solving the characteristic equation $0 = |M - \lambda I|$.
- Find the eigenspace of λ_i by finding the nullspace of $M - \lambda_i I$.

Practice

We asked earlier what the eigenvectors of the rotation matrix $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ are? Likely you said that as it rotates all vectors by 90 degrees, and this not a scale, there are none, so no eigenvalues either. Use the characteristic polynomial of R to find its eigenvalues, (They exist!) and their eigenvectors.

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When the matrix M has dimension greater than 3, the characteristic polynomial has degree greater than three, and finding its roots is usually hard to do. We will have to develop a different way to find the eigens for larger matrices. But before we go onto this, we point out a few larger matrices for which the above approach works, or for which we can use ad hoc reasoning.

Example 5.3. If M is singular, so the nullspace $N(M)$ is more than just $\{0\}$, any vector v in $N(M)$ is an eigenvector with eigenvalue 0. The dimension of the eigenspace for 0 is the nullity of M.

Example 5.4. If P is the projection matrix projecting on the space $V \subset R^n$, then all vectors in V are eigenvectors with eigenvalue 1. Any vector orthogonal to V is an eigenvector with eigenvalue 0.

Example 5.5. The eigenvalues of a triangular matrix are the diagonal entries. Indeed, if say $M = \begin{bmatrix} 1 & 1 & 2 & 3 \\ & 4 & 2 & 3 \\ & & -2 & 1 \\ & & & 4 \end{bmatrix}$ then

$$|M - \lambda I| = \begin{bmatrix} 1-\lambda & 1 & 2 & 3 \\ & 4-\lambda & 2 & 3 \\ & & -2-\lambda & 1 \\ & & & 4-\lambda \end{bmatrix} = (1-\lambda)(4-\lambda)(-2-\lambda)(4-\lambda),$$

which has roots $1, 4, -2$ and 4.

In this previous example, 4 was a root of the characteristic equation of *multiplicity* two. As the characteristic polynomial of an $n \times n$ matrix has degree n there are exactly n eigenvalues, with multiplicity. (But some might not be real.) We will see that an eigenvalue having multiplicity d means that the corresponding eigenspace has dimension at most d.

As the eigenvalues of a triangular matrix are easy to determine, one might try to use Gaussian elimination to triangulise the matrix and then find the eigen values. Unfortunately, elimination does not preserve eigenvalues– if it did, then every non-singular matrix would have the same eigenvalues as the identity. We will look at a different kind of 'diagonalisation' that preserves eigenvalues.

Practice

Show that $|M|$ is the product of the eigenvalues of a matrix M. Show that the sum of the eigenvalues of M is the sum of the diagonal entries.

Problems from the text

5.1: 1, 3, 6, 7, 9, 10, 12, 17, 18, 24, 28 (Non International Student Edition: 1, 3, 8, 5, 11, 9, 13, 19, 20, 27, 26)

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5.2 Diagonalising a matrix

Let S be an $n \times n$ matrix whose columns are eigenvectors of some $n \times n$ matrix M. We get

$$MS = M\left[\begin{array}{c|c|c|c} x_1 & x_2 & \dots & x_n \end{array}\right] = \left[\begin{array}{c|c|c|c} \lambda_1 x_1 & \lambda_2 x_2 & \dots & \lambda_n x_n \end{array}\right] = S\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \dots & \\ & & & \lambda_n \end{bmatrix},$$

where λ_i is the eigenvalue for the vector x_i.

Writing Λ for the diagonal matrix of eigenvalues on the right, we write this as

$$MS = S\Lambda.$$

If the eigenvectors are independent, then S is invertible and so we have the following decomposition

$$S^{-1}MS = \Lambda$$

which because Λ is diagonal, we call a *diagonalisation of* M. We also say that S diagonalises M (by conjugation). A fast and easy application of such a diagonalisation is that is allows us to quickly compute M^d.

Example 5.6. Where $M = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right]$, one can show that $M = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^{-1}$. So

$$M^5 = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]^5\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^{-1} = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\begin{bmatrix} 1^5 & 0 \\ 0 & 3^5 \end{bmatrix}\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^{-1}.$$

The diagonalisation of a matrix, if it exists, need not be unique, Indeed, in our initial example we can re-order the columns of S or multpliy them by constants, and get a different matrix S that also diagonalises M. But as the ith column of MS is λ_i times the ith column of S, the columns of S must be eigenvalues of M.

The multiplicity d of an eigenvalue λ of M as a root of the characteristic polynomial is its *algebraic dimension*. The dimension of its eigenspace is its *geometric dimension*. For M to be diagonalisable, we need that any eigenvalue λ of M of algebraic dimension d has d independent eigenvectors, so has geometric dimension d as well.

Lets observe a couple more properties of eigenvalues. You will be asked to prove simple versions of them.

i. An eigenvector x of M is an eigenvector of M^d (though the converse need not be true) and if its eigenvalue is λ for M then its eigenvalue is λ^d for M^d.

ii. Diagonalisable matrices A and B commute if and only if they are diagonalised by the same matrix S.

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Practice

Assuming that M is diagonalisable, show property (i).

Practice

Show property (ii), in the case that all eigenvalues of A are distinct.

Problems from the text

5.2: 1, 4, 8, 11, 14, 20, 23, 27, 28, 31, 44 (3, 7, 5, 13, 12, 16, 21, 25, 27, 32, 45)

5.3 Difference Equations and Powers M^k

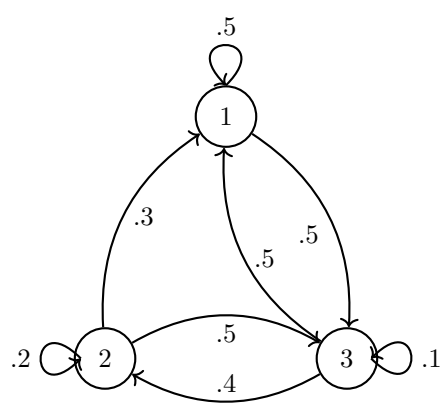
We look at two example applications of eigenvalue and eigenvectors to the field of stochastic processes. In the first we see a Markov process, in the second, the stochastic process is described by a difference equation.

5.3.1 Markov Process

In a Markov process we have a set of n positions, or states, and we let the *state vector*

$$u_t = \begin{bmatrix} p_1(t) \\ p_2(t) \\ \vdots \\ p_n(t) \end{bmatrix}$$

list the probability $p_i(t)$ of being in state i at time t. There is also a rule that tells us the probability of moving from state i to state j in any time-step. For a process with three states, it might look like the following:



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But usually it is easier to represent it by a *transition matrix* T such that $Tu_t = u_{t+1}$:

$$\begin{bmatrix} .5 & .3 & .5 \\ 0 & .2 & .4 \\ .5 & .5 & .1 \end{bmatrix} \begin{bmatrix} p_1(t) \\ p_2(t) \\ p_3(t) \end{bmatrix} = \begin{bmatrix} p_1(t+1) \\ p_2(t+1) \\ p_3(t+1) \end{bmatrix}$$

Notice then that $u_t = T^t u_0$ so the whole system is described by T and u_0. The entries of T are non-negative and the columns of each sum to 1. A matrix for which this holds is called *Markov*. In applications we are interested in what happens to u_t as t changes: does the probabillity of being in a certain state go to 1, and the other states 0? Does the probability of being in a state oscillate? converge? This asymptotic behaviour of the system is related to the eigenvalues of T.

Using a computer, we have computed the eigenvalues of the matrix T given above:

$$\begin{aligned} \lambda_1 &= -2/3 & v_1 &= (1, 2, -3) \\ \lambda_2 &= 1/5 & v_2 &= (1, -1, 0) \\ \lambda_3 &= 1 & v_3 &= (1, 5/13, 10/13) = \frac{1}{28}(13, 5, 10) \end{aligned}$$

Nice! We got an eigenvalue of 1. And what is more, the eigenvector v_3 corresponding to the eigenvalue 1, has all positive entries. These are some useful properties that we will talk about later, but occur more often than you might expect for a Markov matrix. We have scaled the eigenvector so that its entries sum to 1, so can be viewed as probabilities.

If we set this as our position vector:

$$u_t = \frac{1}{28} \begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix}$$

then $u_{t+1} = Tu_t = u_t$. That is, the probability of being in a given state is the same for every time t. This state is called a *steady state*.

But what happens if we start at some other state $u_0 \neq v_3$? To see, we first diagonalise T:

$$T = S\Lambda S^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 5/13 & -1 & 2 \\ 10/13 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1/5 & \\ & & -2/3 \end{bmatrix} \begin{bmatrix} 13/28 & 13/28 & 13/28 \\ 5/12 & -7/12 & -1/4 \\ 5/42 & 5/42 & -3/14 \end{bmatrix}.$$

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We then observe that $u_t = T^t u_0 = S\Lambda^t S^{-1} u_0$. Expanding this we get

$$\begin{aligned}
u_t &= \begin{bmatrix} 1 & 1 & 1 \\ 5/13 & -1 & 2 \\ 10/13 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1/5 & \\ & & -2/3 \end{bmatrix}^t \begin{bmatrix} 13/28 & 13/28 & 13/28 \\ 5/12 & -7/12 & -1/4 \\ 5/42 & 5/42 & -3/14 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \\
&= \underbrace{(\frac{13}{28}p_1 + \frac{13}{28}p_2 + \frac{13}{28}p_3)}_{c_1} 1^t \begin{bmatrix} 1 \\ 5/13 \\ 10/13 \end{bmatrix} \\
&+ (\frac{5}{12}p_1 - \frac{7}{12}p_2 - \frac{1}{4}p_3)(1/5)^t \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \\
&+ (\frac{5}{42}p_1 - \frac{5}{42}p_2 - \frac{3}{14}p_3)(-2/3)^t \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}.
\end{aligned}$$

Now as t gets big, the last two terms disappear, and so

$$u_t \to c_1 \begin{bmatrix} 1 \\ 5/13 \\ 10/13 \end{bmatrix} = \frac{1}{28} \begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix}$$

which is our steady state. This is for any initial position u_0.

This is not the whole situation, but it is almost.

Recall a square matrix is Markov if it has non-negative entries and all columns sum to 1. The following is known as the Perron-Frobenius Theorem.

Theorem 5.7. *If T is a Markov matrix then it has an eigenvalue of 1 and no eigenvalue has greater absolute value. The eigenvector of the eigenvalue 1 can be taken to have no negative entries. If T, or T^k for some positive k has all positive entries, then there is only one eigenvalue with absolute value 1 and it can be taken to have all positive entries.*

Note

The fact that T has a eigenvalue of 1 is easy: as the columns sum to 1, the columns of $T - I$ sum to 0 so are dependent. Thus $\det(T - I) = 0$. The proof of the rest of the theorem is a bit difficult– the proof in the text incomprehensible.

In the case that T has only one eigenvalue of absolute value $\lambda_1 = 1$, then $u_t \to cv_1$ for some constant c. If there are more eigenvalues with absolute value 1, then the situation is a little more complicated. We will discuss this a little more later.

Example 5.8. Each year $\frac{1}{10}$ of the people in Daegu move to Ulsan, and $\frac{2}{10}$ of the people in Ulsan move to Daegu. If there are 1000 people in Korea, (and obviously that all live in one of these two cities,) then what are the stable populations of the two cities? (It is not hard to see that $666\frac{2}{3}$ people live in Daegu and $333\frac{1}{3}$ in Ulsan, but... we use matrices.)

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Where the state vector $u_t = \begin{bmatrix} D, U \end{bmatrix}$ tells us the respective proportions D and U of the population that live in Daegu and Ulsan, the transition matrix is the Markov matrix

$$T = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix}.$$

We know it has only one eigenvalue $\lambda_1 = 1$ with absolute value 1, and the eigenvector for this is the solution x_1 to:

$$0 = \begin{bmatrix} .9-1 & .2 \\ .1 & .8-1 \end{bmatrix} x = \begin{bmatrix} -.1 & .2 \\ .1 & -.2 \end{bmatrix} x.$$

Thus $x_1 = (2, 1)$. Scaling so that the total population is 1000, we get $D = 666\frac{2}{3}$ people living in Daegu, and $333\frac{1}{3}$ in Ulsan.

5.3.2 Difference Equations

Consider a recursive description of a population $p(t)$ at time t such as

$$p(0) = 100 \qquad \text{and} \qquad p(t) = p(t-1) \cdot 1.2.$$

We can solve this as

$$p(t) = 100(1.2)^t.$$

This is easy, but it gets more difficult when the recurrence is more complicated.

Example 5.9. The fibonacci recurrence for describing a population of rabbits is

$$F_0 = 0, F_1 = 1, F_2 = 1, \qquad \text{and} \qquad F_{t+2} = F_{t+1} + F_t \text{ for } t \geq 2.$$

To solve this (to find a closed formula for F_t) we look at it as a stochastic process with

$$u_{t+1} = \begin{bmatrix} F_{t+2} \\ F_{t+1} \end{bmatrix} \text{ depending on } u_t = \begin{bmatrix} F_{t+1} \\ F_t \end{bmatrix}.$$

The transmission matrix T should yield

$$T \begin{bmatrix} F_{t+1} \\ F_t \end{bmatrix} = Tu_t = u_{t+1} = \begin{bmatrix} F_{t+2} \\ F_{t+1} \end{bmatrix} = \begin{bmatrix} F_{t+1} + F_t \\ F_{t+1} \end{bmatrix},$$

and so we see that $T = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Now this is not a Markov matrix, so we do not know for sure that there is an eigenvalue of 1, but lets find the eigenvalues anyways.

Solving

$$0 = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1 = (\lambda - 1/2)^2 - 5/4$$

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gives $\lambda = \frac{1\pm\sqrt{5}}{2}$. From the second row we get that the eigenvector x_i for λ_i is $(\lambda_i, 1)$. So where $u_0 = (F_1, F_0) = (1, 0)$, we have

$\lambda_1 - \lambda_2 = \frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = \sqrt{5}$

$$\begin{aligned}
u_t &= T^t u_0 = S\Lambda^t S^{-1} u_0 = \left[\begin{smallmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{smallmatrix}\right] \begin{bmatrix} \lambda_1^t & \\ & \lambda_2^t \end{bmatrix} \left[\begin{smallmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{smallmatrix}\right] \frac{(1,0)}{\lambda_1 - \lambda_2} \\
&= \frac{(1, -\lambda_2)\cdot(1,0)}{\sqrt{5}} \lambda_1^t \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} + \frac{(-1, \lambda_1)\cdot(1,0)}{\sqrt{5}} \lambda_2^t \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \\
&= \frac{1}{\sqrt{5}} \lambda_1^t \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{5}} \lambda_2^t \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \\
&= \frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_1^{t+1} - \lambda_2^{t+1} \\ \lambda_1^t - \lambda_2^t \end{bmatrix}.
\end{aligned}$$

The bottom row of this is

$$F_t = \frac{\lambda_1^t - \lambda_2^t}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^t - \left(\frac{1-\sqrt{5}}{2} \right)^t \right) \longrightarrow \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^t$$

as $t \to \infty$.

Note

Actually, the absolute value $\frac{1}{\sqrt{5}} \left| \frac{1-\sqrt{5}}{2} \right|^t$ of the second term is always less than $1/2$, so F_t is just the closest integer to $\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^t$.

In general when solving a recurrence $u_{t+1} = Tu_t$ for T with distinct eigenvalues, each eigenvalue λ_1 of T yields a part

$$c_1 \lambda_i^t x_i$$

of the solution. The full solution is of the form

$$u_t = c_1 \lambda_1^t x_1 + c_2 \lambda_2^t x_2 + \cdots + c_n \lambda_n^t x_n.$$

Often, the eigenvalues are ordered so that

$$\lambda_n \le \cdots \le \lambda_2 \le \lambda_1.$$

Then λ_1 is the maximum eigenvalue and the difference equation is unstable if $|\lambda_1| > 1$.

As t gets large, the eigenvalues of maximum absolute value dominate, and the other terms can be thrown away.

The difference equation is...

- *stable* if $|\lambda_i| < 1$ for all i, the limiting value is 0.
- *neutrally stable* if $|\lambda_i| = 1$ for some i and $|\lambda_j| < 1$ for all $j \neq i$. The llmiting state may be $c_i x_i$, or bounce between $-c_i x_i$ and $c_i x_i$, or if λ_i is complex, it may have recur with some longer period.
- *unstable* if there is an eigenvalue with $|\lambda_i| > 1$, the state grows infinitly.

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Note

What if two eigenvalues have absolute value 1? Is the state not also stable then? Not necessarily, see the text for an example. What if two eigenvalues are 1?

Problems from the text

5.3: 1, 3, 4, 5, 6, 8, 20, 23, 27 (2, 1, 5, 4, 8, 7, 22, 23, 25)

5.4 Differential Equations

A system of differential equations is simply a system of equations involving derivatives. Solving one means finding the functions for which the equations hold.

Example 5.10. Solve the differential equation: $v'(t) = a \cdot v(t)$. Perhaps we remember that $v(t) = e^{at}$ is a solution to this equation, as $\frac{\mathrm{d}}{\mathrm{d}t} e^{at} = ae^{at}$.

As $v(t) = ce^{at}$ is also a solution, there is usually and extra 'initial value' needed to determine a particular solution.

Practice

Find $v(t)$ when $v'(t) = av(t)$ and $v(0) = 4$.

This becomes more difficult with more variables.

Example 5.11. Solve

$$\begin{aligned} \frac{\mathrm{d}v}{\mathrm{d}t} &= 4v - 5w \text{ where} \qquad & v(0) = 8 \\ \frac{\mathrm{d}w}{\mathrm{d}t} &= 2v - 3w \text{ where} \qquad & w(0) = 5 \end{aligned}$$

To solve this, we will first guess that the solution is of the form

$$v(t) = ae^{\lambda t} \qquad w(t) = be^{\lambda t}$$

for some constants a, b and λ, and then find these constants.

Plugging our guesses into the given equations, we get

$$\begin{aligned} \lambda ae^{\lambda t} &= 4ae^{\lambda t} - 5be^{\lambda t} \\ \lambda be^{\lambda t} &= 2ae^{\lambda t} - 3be^{\lambda t}. \end{aligned}$$

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Defining the matrices

$$u(t) = \begin{bmatrix} v(t) \\ w(t) \end{bmatrix} \qquad T = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$$

this the orignal system can be written as

$$u' = \frac{\mathrm{d}}{\mathrm{d}t} u = Tu \qquad u(0) = (8, 5) \tag{1}$$

where $u' = \frac{\mathrm{d}}{\mathrm{d}t} u$ is the obvious shorthand for $(v'(t), w'(t))$. With our guessed solution plugged in, this becomes

$$e^{\lambda t} \lambda \begin{bmatrix} a \\ b \end{bmatrix} = e^{\lambda t} T \begin{bmatrix} a \\ b \end{bmatrix} \tag{2}$$

which has the same solutions as $T\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right] = \lambda\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right]$. So to find a, b and λ we have to find the eigens of T.

You can do this:

$$\begin{aligned} \lambda_1 &= -1 & x_1 &= (1, 1) \\ \lambda_2 &= 2 & x_2 &= (5, 2) \end{aligned}$$

Each pair (λ_i, x_i) gives a solutions to the system $u' = Tu$, so we have two:

$$\begin{aligned} u(t) &= e^{\lambda_1 t} x_1 = e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ u(t) &= e^{\lambda_2 t} x_2 = e^{2t} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \end{aligned}$$

and as a linear combination of solutions is a solution, we have a solution space

$$u(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} e^{-t} & \\ & e^{2t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Our initial conditions $u(0) = (5, 8)$ designated one solution:

$$\begin{bmatrix} 8 \\ 5 \end{bmatrix} = u(0) = \underbrace{\begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} e^0 & \\ & e^0 \end{bmatrix}}_{e^{\Lambda t}} \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{S^{-1}u(0)} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

which we can solve:

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 5 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} 2 & -1 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 5 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} -9 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Thus $u(t) = 3e^{-t}\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right] + e^{2t}\left[\begin{smallmatrix} 5 \\ 2 \end{smallmatrix}\right]$ is our solution.

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Note

Summarising: the differential equation $\frac{\mathrm{d}}{\mathrm{d}t}u = Tu$ has the solution

$$u(t) = Se^{\Lambda t}S^{-1}u(0)$$

where T is diagonalised $T = S\Lambda S^{-1}$.

5.4.1 Matrix exponential

We cheated with the notation a bit above, writing

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & e^{\lambda_d t} \end{bmatrix}.$$

Lets define this properly. Recall that the McClauren series for e^x is

$$e^x = 1 + x + x^2/2! + x^3/3! + \dots$$

In analogue to this we defined, for a matrix M,

$$e^{Mt} = I + Mt + (Mt)^2/2! + (Mt)^3/3! + \dots.$$

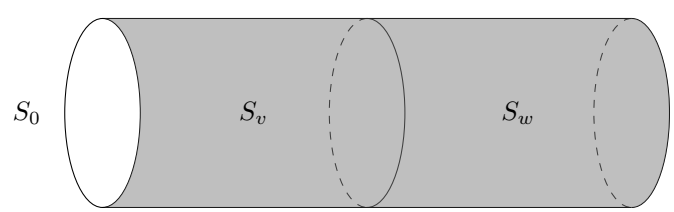
This agrees with our earlier use of the notation, and also has some properties that are envoked by the notation:

- $e^{Ms}e^{Mt} = e^{M(s+t)}$
- $e^{Mt}e^{-Mt} = I$
- $\frac{\mathrm{d}}{\mathrm{d}t}e^{Mt} = Me^{Mt}$.

Again, in this last, the notation $\frac{\mathrm{d}}{\mathrm{d}t}M$ means that we take a derivative entrywise. Thus e^{Mt} is the solution to $u'(t) = Mu(t)$, and in fact $e^{Mt} = Se^{\Lambda t}S^{-1}$.

5.4.2 A physical example

Differential equations can arise from looking at the movement of gas in a pipe.



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In two regions S_v and S_w, we start at time $t = 0$ with concentrations $v(0)$ and $w(0)$. In the region S_0 outside the pipe, we assume the concentration of gas is essentially 0. At time t gas moves between regions proportionally to the difference in concentrations. So the flow is

$$\frac{\mathrm{d}v}{\mathrm{d}t} = (0 - v) + (w - v) \qquad \frac{\mathrm{d}w}{\mathrm{d}t} = (0 - w) + (v - w).$$

To find $v(t)$ and $w(t)$ we solve $u' = \left[\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right] u$. We start be diagonalising the transition matrix $T = \left[\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right]$:

$$T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & \\ & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$$

$$\begin{aligned} u(t) &= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^{-t} & \\ & e^{-3t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1} u(0) \\ &= c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

where $\left[\begin{smallmatrix} c_1 \\ c_2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]^{-1} u(0)$.

5.4.3 Stability

For the equation $u' = Tu$ the solution is of the form

$$u(t) = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2 + \cdots + c_n e^{\lambda_n t} x_n$$

where $T^{-1}u(0) = [c_1, \ldots, c_n]^T$. This is stable if $e^{\lambda_i t} \to 0$ for all i and (usually) blows up if any of the $e^{\lambda_i t}$ go to infinity. Observing that

$$|e^{\lambda t}| = e^{at} \text{ where } \lambda = a + bi$$

we see that $e^{\lambda_i t}$ goes to infinity iff the real part $Re(\lambda)$ of λ is greater than 0.

Note

The system $u' = Tu$ is...

- *stable* if $Re(\lambda_i) < 0$ for all i.
- *neutrally stable* if $Re(\lambda_i) \leq 0$ for all i and $Re(\lambda_j) = 0$ for some j.
- *unstable* if $Re(\lambda_i) > 0$ for some i.

Problems from the text

5.4: 1, 2, 3, 6, 7, 10, 16, 18 (1, 2, 3, 9, 8, 5, 20, 19)

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5.5 Complex Matrices

Okay, complex numbers are sneaking into our examples. The picture is incomplete without them. Let's look at matrices over the field $\mathbb{C}$ of complex numbers. This requires us to generalise several definitions such as length and inner (dot) product.

5.5.1 Complex Numbers

Recal lthat complex numbers are of the form

$$z = a + ib$$

where $a, b \in \mathbb{R}$ and $i^2 = -1$. So arithmetic works as expected:

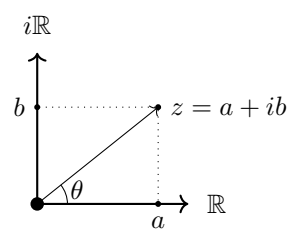
- $(a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$, and
- $(a_1 + ib_1)(a_2 + ib_2) = (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)$.

For a complex number $z = a + ib$, the value $a = Re(z)$ is the *real part* of z and $b = Im(z)$ is the *imaginary part*.

Multiplication is easier to deal with if we use polar notation. Recall, in polar notation, that the complex number $z = a + ib$ can be written as

$$re^{i\theta} = r\sin\theta + ir\cos\theta$$

where $r = \sqrt{a^2 + b^2}$ is length of z and $\theta = \arctan(b/a)$ (for $a \geq 0$) is the angle that z makes with the real axis when we graph it in the complex plane:



Now, multiplication works as:

$$r_1e^{i\theta_1} \cdot r_2e^{i\theta_2} = (r_1 \cdot r_2)e^{i(\theta_1+\theta_2)}.$$

Moreover, this length r, this is what we use as the length $|z|$ of the z. A complex number $z = a + ib$ is real if $b = 0$, and in this case $|z| = \sqrt{a^2 + b^2} = \sqrt{a^2} = |a|$, so this generalises our notion of absolute value for real numbers.

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5.5.2 Complex vectors: Length and inner product

Let $x = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}$ be in $\mathbb{C}^n$. The *length* or *norm* $\|x\|$ is defined by

$$\|x\|^2 = |z_1|^2 + \cdots + |z_n|^2.$$

Practice

What is the length of the complex vector $x = (2, 1 + 2i)$?

The inner product for real vectors was

$$x \cdot y = x^T y = x_1 \cdot y_1 + \cdots + x_n \cdot y_n.$$

We could use this same formula for complex numbers, but then we would loose such properties as

$$x \cdot x = \|x\|^2.$$

Practice

Give an example showing this.

To define a proper inner product, we first need to define a 'complex conjugate'. For $z = a + ib = re^{i\theta}$, the *complex conjugate* of z is the number

$$\overline{z} = a - ib = re^{-i\theta}.$$

Practice

Show that $z\overline{z} = |z|^2$.

Now, for complex (column) vectors $x = (x_1, \ldots, x_n)^T$ and $y = (y_1, \ldots, y_n)^T$ we let $\overline{x} = (\overline{x}_1, \ldots, \overline{x}_n)^T$ and define the inner product as

$$x \cdot y = \overline{x}^T y = \overline{x}_1 \cdot y_1 + \cdots + \overline{x}_n \cdot y_n.$$

It follows then that

$$\begin{aligned} x \cdot x = \overline{x}^T x &= \overline{x}_1 \cdot x_1 + \cdots + \overline{x}_n \cdot x_n \\ &= |x_1|^2 + \cdots + |x_n|^2 = \|x\|. \end{aligned}$$

5.5.3 The Hermitian

The *(complex) conjugate* of a matrix $M = [m_{ij}]$ is the matrix

$$\overline{M} = [\overline{m}_{ij}]$$

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we get from M by entrywise conjugation.

So, for example

$$\overline{\begin{bmatrix} 1 & 4 \\ 1+i & i \end{bmatrix}} = \begin{bmatrix} 1 & 4 \\ 1-i & -i \end{bmatrix}.$$

The *Hermitian* or *conjugate transpose* of M is

$$M^H := \overline{M}^T = \overline{M^T}.$$

Practice

What is the Hermitian of $\left[\begin{smallmatrix} 2+i & 4 \\ i & 0 \\ 1 & 2-i \end{smallmatrix}\right]$?

Observe that

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}^H = [\overline{x}_1, \ldots, \overline{x}_n],$$

so the inner product can now be written as $x \cdot y = x^H y$.

Though $x^T = x^H$ when x is a real vector, so $x^T x = x^H x$, this is not true for complex vectors x. In many situations, when we are considering complex vectors, we must use x^H instead of x^T in results or definition we know from the real case. One such situation is the definition of orthogonality. From now on:

Note

Two vectors x and y are *orthogonal* if $x^H y = 0$.

Observe also, that much like we had for the transpose, we have

$$(AB)^H = \overline{(AB)^T} = \overline{(B^T A^T)} = \overline{B^T}\ \overline{A^T} = B^H A^H.$$

Practice

Why does that second last equality hold?

A matrix M is *Hermitian* if

$$M^H = M.$$

For example $\left[\begin{smallmatrix} 2 & 3-i \\ 3+i & 5 \end{smallmatrix}\right]$ is Hermitian. One quickly sees that for a Hermitian matrix $M = [m_{ij}]$ we must have

- m_{ii} is real, and
- $\overline{m}_{ij} = m_{ji}$.

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Hermitian matrices have three nice properties, and keep in mind, **if a matrix is real, it is Hermitian if and only if it is symmetric,** and so these properties hold for real symmetric matrices too. Let M be Hermitian.

i. For all $x \in \mathbb{C}^n$, $x^H M x$ is real.

ii. All eigenvalues of M are real.

iii. Eigenvectors of M with distinct eigenvalues are orthogonal.

As a hermitian matrix M has distinct eigenvalues, we can diagonalise it as

$$M = S\Lambda S^{-1}.$$

The orthogonality of the eigenvectors means that if normalised, then $SS^H = I$, and so we get

$$M = S\Lambda S^H,$$

which is nice as S^H is much easier to compute than S^{-1} is in general.

Note

We will show in the next section that this result actually holds for any Hermitian matrix M, not only those with distinct eigenvalues. This will come down to showing that for a Hermitian matrix, every eigenvalue has the same geometric mutliplicity as algebraic multiplicity.

Proof of properties i. ii. and iii.

To see property i., observe that the 1×1 matrix $x^H M x$ is its own conjugate:

$$(x^H M x)^H = x^H M^H x^{HH} = x^H M^H x = x^H M x$$

and so must be real.

To see property ii., let $Mx = \lambda x$. Then

$$\underbrace{x^H M x}_{\text{real by i.}} = x^H \lambda x = \lambda x^H x = \lambda \|x\|$$

and $\|x\|$ is real (and non-zero) so $\lambda = \frac{x^H M x}{\|x\|}$ is real.

To see property iii., let

$$Mx = \lambda_1 x \text{ and } My = \lambda_2 y.$$

Then

$$\lambda_1 x^H y = (Mx)^H y = x^H M y = x^H \lambda_2 y = \lambda_2 x^H y,$$

so $\lambda_1 \neq \lambda_2$ gives that $x^H y = 0$, as needed.

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5.5.4 Unitary Matrices

Generalising on the idea of an orthogonal matrix, a complex matrix U is *unitary* if

$$U^H U = I.$$

Practice

Show the following properties for any unitary matrix U.

i. $(Ux)^H(Uy) = x^H y$ and in particular $\|Ux\| = \|x\|$.

ii. Every eigenvalue λ of U has $|\lambda| = 1$.

iii. Eigenvectors of different eigenvalues are orthonormal.

As we mentioned above, we will show in the next section that every Hermitian matrix will be diagonalisable by a unitary matrix. Let's see an example of this.

Example 5.12. The matrix $M = \left[\begin{smallmatrix} 3 & 2+2i \\ 2-2i & 1 \end{smallmatrix}\right]$ is Hermitian. Let's diagonalise it.

$$\begin{aligned} Det(M - \lambda I) &= (3-\lambda)(1-\lambda) - (2+2i)(2-2i) = (3-\lambda)(1-\lambda) - 8 \\ &= \lambda^2 - 4\lambda - 5 = (\lambda - 2)^2 - 9 \end{aligned}$$

has roots $\lambda = 2 \pm \sqrt{9} = -1, 5$. To get the eigenvector for $\lambda_1 = 5$ we find the null-vector of $\left[\begin{smallmatrix} -2 & 2+2i \\ 2-2i & -4 \end{smallmatrix}\right]$ which is $x_1 = (1+i, 1)$, and read off that the eigenvector of λ_2 is $x_2 = (1+i, -2)$. These should be orthogonal:

$$(1-i, 1) \cdot (1+i, -2) = (1-i)(1+i) + (1)(-2) = 2 - 2 = 0.$$

Normalising them to $x_1 = (1+i, 1)/\sqrt{3}$ and $x_2 = (1+i, -2)/\sqrt{6} = (1, i-1)/\sqrt{3}$ we get that $U = \frac{1}{\sqrt{3}}\left[\begin{smallmatrix} 1+i & 1 \\ 1 & -1+i \end{smallmatrix}\right]$ and so $U^{-1} = U^H = \frac{1}{\sqrt{3}}\left[\begin{smallmatrix} 1-i & 1 \\ 1 & -1-i \end{smallmatrix}\right]$.

We should check that $M = U \begin{bmatrix} -1 & \\ & 5 \end{bmatrix} U^{-1}$.

Problems from the text

5.5: 1, 2, 6, 10, 12, 15, 18, 29, 37, 42, 43, 49, 50 (1, 7, 2, 6, 18, 17, 12, 30, 40, 37, 50, 41, 36)

5.6 Similarity Transformations

A matrix B is similar to a matrix A if

$$B = MAM^{-1}$$

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for some matrix M.

Similar matrics share similar properties For example, if $B = MAM^{-1}$ then A and B have the same eigenvalues, and their eigenvectors are related through M:

$$\begin{aligned} Ax = \lambda x \quad &\iff \quad M^{-1}BMx = \lambda x \\ &\iff \quad B(Mx) = \lambda(Mx) \end{aligned}$$

Practice

Show that if B is similar to a diagonalisable matrix, then B is diagonalisable.

Indeed, that a matrix is diagonalisable just means that it is similar to a diagonal matrix. Recalling our long goal of computing the eigenvalues of a big matrix, this suggests where we might be going. If we can find a diagonalisation of a matrix M then we have its eigenvalues. For this goal, it enough that M is similar to a triangular matrix. But we will see in the next subsection why it is nicer if M is similar to a diagonal matrix. We will see also that while similarity is nice, similary by unitary matrices is nicer. So the ideal will be writing $M = U\Lambda U^{-1}$ for some unitary U and diagonal Λ. We will look at which matrices we can do this for. It won't be all. For the other matrices we will show that they can be triangulised, but better than this, triangulised by unitary matrices. And that they can also by 'almost diagonalised', into something we will call theIR Jordan form.

First, we look, from a new point of view, at why diagonalising is nicer than triangulating, if we can do it.

5.6.1 Change of Basis

Given a basis $B = \{b_1, \ldots, b_n\}$ and a vector x the notation

$$[x]_B = (a_1, \ldots, a_n)$$

tells us that $x = a_1b_1 + a_2b_2 + \cdots + a_nb_n$.

Example 5.13. Where $E = \{e_1, e_2\}$ is the standard basis $[(1,3)]_E = (1,3)$ but where B is the basis $B = \{b_1 = (1,1), b_2 = (-1,1)\}$ we have $[(1,3)]_B = (2,1)$ because $(1,3) = 2(1,1) + 1(-1,1)$.

Given a vector $[x]_B$ represented in the basis B, it is easy to transform it to the standard basis E, but how do we find $[x]_B$ for a given vector x, as we

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did in the above example? Well, the vector $b_1 = \left[\begin{smallmatrix}1\\1\end{smallmatrix}\right]$ is expressed in basis B as $[b_1]_B = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$. This can be written as

$$\begin{bmatrix}1\\1\end{bmatrix} = M_B \begin{bmatrix}1\\0\end{bmatrix}$$

where M_B is the matrix whose i^{th} column is the ith vector b_i in B. More generally we have that if $[x]_b = (a_1, \ldots, a_n)$, so $x = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$, then

$$x = M_B [x]_B.$$

So to go in the other direction, to find the representation $[x]_B$ of the vector x in the basis B, we compute

$$[x]_B = M_B^{-1} x.$$

Example 5.14. To represent $x = (1, 3)$ in the basis $B = \{(1, 1), (-1, 1)\}$, we find $M_B^{-1} = \left[\begin{smallmatrix}1 & 1\\-1 & 1\end{smallmatrix}\right]/2$ and compute

$$[(1, 3)]_B = M_B^{-1} x = \frac{1}{2}\begin{bmatrix}1 & 1\\-1 & 1\end{bmatrix}\begin{bmatrix}1\\3\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}.$$

Note:

Change of basis can be confusing. We talk of the vectors x and $M_B^{-1} x$ as being the same vector, just represented in different bases. But hey, multiplying by the transformation M_B^{-1} takes the vector $(1, 3)$ to $(2, 1)$! It does change. Yep, confusing. The point of view of the change of basis is that we are keeping the vectors still, but moving the paper (and the coordinate axes). Instead of rotating vectors clockwise, we rotate the paper counterclockwise. Instead of stretching the vectors, we zoom out the paper. Multiplying by M_B changes from the B frame of reference to the standard E frame of reference. (Notation can be the other way around in other books).

Example 5.15. The matrix $A = \left[\begin{smallmatrix}3 & 1\\1 & 3\end{smallmatrix}\right]$ diagonalises to

$$A = M \Lambda M^{-1} = \begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}4 & \\ & 2\end{bmatrix}\begin{bmatrix}1 & 1\\-1 & 1\end{bmatrix}/2.$$

Applying this as a transformation to some vector $x = (1, 3)$ we get

$$A : (1, 3) \xmapsto{M_B^{-1}} (2, 1) \xmapsto{\Lambda} (8, 2) \xmapsto{M_B} (6, 10).$$

We interpret transformation by A as consisting of three steps. The first is changing $x = (1, 3)$ to $[x]_B = (2, 1)$ in the basis B. Then in the basis B transforming by Λ to $(8, 2)$. Transformations by diagonal matrices are easy to

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visualise as there is no rotation involved, just stretching in the two directions. Finally, we change back to $(6, 10)$ in the original basis.

So a matrix A being similar to a diagonal matrix is pretty nice. It is even nicer if the matrix M that transforms A to diagonal is unitary (orthanormal). Not only is M^{-1} then easy to compute, but if M is unitary, then the eigenspaces of A are orthogonal, so basis we are transforming to is orthonormal. Thus the action of A decomposes into projections onto the eigenspaces:

$$A = U\Lambda U^H = \lambda_1 x_1 x_1^H + \cdots + \lambda_n x_n x_n^H.$$

Example 5.16. Where

$$\begin{aligned} A = \frac{1}{2}\begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} &= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= 2\begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 1\begin{bmatrix} 1/2 & -1/2 & \\ -1/2 & 1/2 & \\ & & \end{bmatrix} + 1\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

The first matrix on the left of the last line is the projection onto the eigenspace $\langle(1, 1, 0)\rangle$ of $\lambda_1 = 2$ while the second two matrices are the projection onto the eigenspace $\langle(1, -1, 0), (0, 0, 1)\rangle$ of $\lambda_2 = 1$.

5.6.2 Triangulating by Unitary Matrices

We cannot always diagonalise a matrix by a unitary matrix, but we can at least triangulate it. The following is called Schur's Lemma.

Theorem 5.17. *Any matrix A can be triangulated: $A = UTU^H$ by a unitary matrix U. The eigenvalues of A are on the diagonal of T.*

Proof. If $A = UTU^H$ for triangular T the fact that the diagonals of T are the eigenvalues of A is immediate from the fact that similar matrices share eigenvalues. So it is enough to prove the first statement of the theorem. See the text for this. □

With this we can now answer the question of which matrices A can be diagonalised by unitary matrices. A matrix A is *normal* if $A^H A = AA^H$.

Practice

Show that $(A^H A)^H = A^H A$ for any matrix A.

Corollary 5.18. *The following are equivalent.*

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i. $A = U\Lambda U^H$ *for some unitary matrix* U*.*

ii. A *has* n *orthogonal eigenvectors.*

iii. A *is normal.*

Proof. The proof of $i. \iff ii.$ is trivial as the columns of such a U must be eigenvectors. The equivalence of $ii.$ and $iii.$ comes from observing that where $A = UTU^H$ we have

$$AA^H = UTT^HU^H \text{ and } A^HA = UT^HTU^H$$

and so

$$(AA^H = A^HA) \iff (TT^H = T^HT).$$

Showing that $TT^H = T^HT$ for triangular T if and only if T is diagonal then yields the result. Certainly if T is diagonal we have $TT^H = T^HT$. Question 21 of the text asks you to show that if $TT^H = T^HT$ for triangular T, then T is diagonal. □

Observe now that hermitian matrices A are normal, so whether or not their eigenvalues are distinct, the are diagonalisable by unitary matrices. We claimed this last section without proof.

5.6.3 Jordon Canonical Form

If we relax the restriction to conjugation by unitary matrices, we can show that a matrix is similar to something even closer to diagonal that a triangular matrix.

An $n \times n$ $M = [m_{ij}]$ matrix is *block diagonal* if the index set $[n]$ can be partitioned into sets $B_1, \ldots, B_d$ of consecutive indices such that m_{ij} is non-zero only if i and j are in the same set B_k for some k. The square submatrices made of the rows and columns in B_k for each k are called *blocks*. A block is *Jordan* if its diagonal entries are all the same, and the only other non-zero entries are ones on the super-diagonal. A matrix is *Jordan* if it is block diagonal with Jordan blocks.

The proof of the following theorem can be found in the appendix of the text.

Theorem 5.19 (Jordon Form)**.** *Any matrix* A *can be written (uniquely) as* $A = MJM^{-1}$ *for a Jordon matrix* J*. (The eigenvalues of* A *are the diagonals of* J*. For this representation to be unique, the diagonals must assumed to be non-decreasing, and for Jordan block with the same diagonals, the small block must be assumed to come first.)*

If $A = MJM^{-1}$ for a jordan matrix J, J is the *Jordan canonical form* of A.

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Example 5.20. The matrices $A = \begin{bmatrix} 11 & -2 & 7 \\ 0 & 8 & 0 \\ 3 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & -2 & -5 \\ 4 & 8 & 4 \\ 1 & -2 & 3 \end{bmatrix}$ both have eigenvalues $4, 8, 8$. For A the eigenspacess have dimensions 1 and 2, for B they have dimensions 1 and 1. So the Jordon forms are

$$J_A = \begin{bmatrix} 4 & & \\ & 8 & \\ & & 8 \end{bmatrix} \qquad \text{and} \qquad J_B = \begin{bmatrix} 4 & & \\ & 8 & 1 \\ & & 8 \end{bmatrix}$$

respectively.

There may be more than one Jordon block for a given eigenvalue, as we see with A above. The number of Jordon blocks for an eigenvalue is determined by the algebraic multiplicity and the geometric multiplicity of that eigenvalue. Knowing that there are two blocks for an eigenvalue does not tell us the size of each.

The following is an easy corollary of the above theorem.

Practice

Show that two matrices are similar if and only if they have the same Jordon Canonical Form.

Not only is the Jordon form useful in that it lets us decide if two matrices are similar, as with a diagonalisation, it is useful for taking powers.

Practice

Show that

$$\begin{bmatrix} a & & \\ & b & 1 \\ & & b \end{bmatrix}^n = \begin{bmatrix} a^n & & \\ & b^n & nb^{n-1} \\ & & b^n \end{bmatrix}$$

and that

$$\begin{bmatrix} a & 1 & \\ & a & 1 \\ & & a \end{bmatrix}^n = \begin{bmatrix} a^n & na^{n-1} & \binom{n}{2}a^{n-1} \\ & a^n & na^{n-1} \\ & & a^n \end{bmatrix}.$$

Problems from the text

5.6: 1, 3, 5, 6, 8, 11, 14, 16, 26, 36, 40 (5, 2, 7, 4, 8, 11, 18, 13, 30, 38, 42)

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6 Positive Definite Matrices

6.1 Maxima, minima, and saddle points

We return to considering only real matrices. This chapter is starts with a simple idea. One can make an n-variate polynomial $f(x_1, \ldots, x_n)$ from an $n \times n$-matrix A by conjugating it by a vector x of variables: $f(x_1, \ldots, x_n) = x^T Ax$. For example, for a 2×2-matrix A we get

$$f(x, y) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + (b + c)xy + dy^2.$$

Observe that replacing each of b and c with $\frac{b+c}{2}$ does not change this polynomial. So if we are interested in representing $f(x, y)$ we could have used a symmetric matrix A instead. It is not hard to see that this holds more generally, for all n. We will therefore restrict our attention to symmetric real matrices. Recall they are nice in that their eigenvalues and eigenvectors are real.

Observe also that every monomial in the polynomial $f(x, y)$ has degree 2. Thus the polynomial is a homogeneous polynomial of degree 2. In particular, it has no constant term, and so we always have $f(0, 0) = 0$.

Practice

Show that this holds more generally in dimension n: if A is $n \times n$ then $x^T Ax$ is an n-variate homogeneous polynomial of degree 2.

As a function $f(x, y)$ is *positive definite* if $f(x, y) > 0$ for all $x, y \in \mathbb{R}^2 \setminus \{(0, 0)\}$, we call a symmetric real matrix A *positive definite*

$$x^T Ax > 0$$

for all non-zero (real!) vectors x.

This means, in particular, that all eigenvalues are positive. Indeed, for a normalised eigenvector x_i

$$x_i^T Ax_i = x_i^T x_i \lambda_i = \lambda_i.$$

We also know that a positive definite matrix must have positive numbers on the diagonal. Indeed, taking $x = e_i$ we get that $x^T Ae$ is the ith diagonal of A.

Are these properties enough to conclude that a symmetric matrix A is positive definite? What conditions guarantee that it it? Well, let's look at this.

We get one set of conditions from calculus. Let's start with 2×2 matrices.

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In general, we have

$$f(x,y) = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2.$$

The critical points occur at (x, y) such that

$$0 = \frac{\mathrm{d}}{\mathrm{d}x} f(x,y) = 2ax + 2by$$
$$0 = \frac{\mathrm{d}}{\mathrm{d}y} f(x,y) = 2bx + 2cy$$

or equivalently, such that

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which we have observed holds for $(x, y) = (0, 0)$.

The set of critical points of $f(x, y)$ are the nullspace of A. If this has dimension 0 then $(0, 0)$ and the graph of $z = f(x, y)$ is either a paraboloid or a saddle point. If it has dimension 1 then it is a line through $(0, 0)$ and the graph of $z = f(x, y)$ is a 'valley'. If it is dimension 2 then $f(x, y) = 0$ everywhere.

How did we differentiate these cases in multivariable calculus? The paraboloid that we are looking for occurs when the critical point $(0, 0)$ is a local minimum. (If it is a local maximum the function is negative definite, if it is neigher, the graph of the function is a saddle point.)

The matrix A is positive definite if and only if $(0, 0)$ is a local minimum. For this to be true, we need that all directional derivatives at $(0, 0)$ are positive. In particular we need that

$$0 < \frac{\mathrm{d}^2}{\mathrm{d}x^2} f(x,y) = 2a$$
$$0 < \frac{\mathrm{d}^2}{\mathrm{d}y^2} f(x,y) = 2c$$

which is exactly the condition that the diagonals are positive.

We also need that the other directional derivatives are positive, and by the second derivative test, this is true if and only if

$$f_{xx} f_{yy} > f_{xy}^2 \qquad \text{i.e. } ac > b^2.$$

In summary,

Note

The symmetric matrix $\left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$ is positive definite if and only if $a > 0$ and $ac > b^2$.

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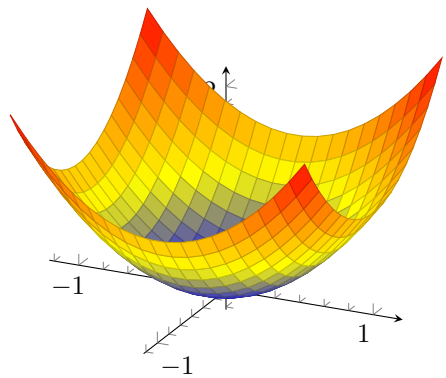
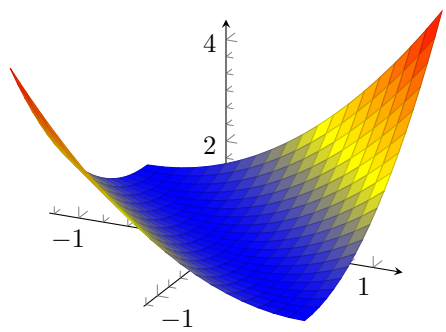
Recall from calculus what $z = f(x, y)$ looks like at $(0, 0)$ in terms of $ac - b^2$.

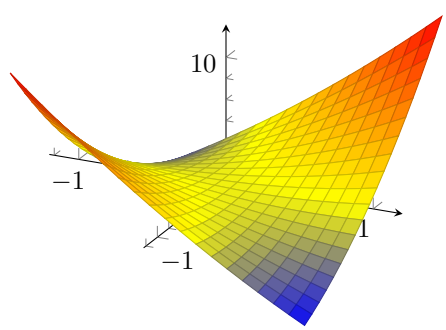
If $ac - b^2 > 0$ then (as we discussed) $(0, 0)$ is a local min and $z = f(x, y)$ is a upward opening parboloid.



If $ac = b^2$ as with the polynomial $(x + y)^2$ then $z = f(x, y)$ is a valley.



If $ac < b^2$ then $(x, y) = (0, 0)$ is a saddle point of $z = f(x, y)$.



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Problems from the text

6.1: 1, 2, 3, 5, 6, 10, 21 (6, 3, 7, 1, 9, 8, 19)

6.2 Tests for Positive Definiteness

In the last section we saw a nice characterisation of the 2×2 symmetric matrices that are positive definiteness. We won't get such an 'equational' characterisation for $n \times n$ positive definite matrices, but we do get several characterisations.

Theorem 6.1. *The following are equivalent definitions for a symmetric matrix A to be positive definite.*

i. $x^T Ax > 0$ *for all* $x \neq 0$.

ii. All eigenvalues of A are positive.

iii. All upper left square submatrices of A have positive determinants.

iv. All pivots of A (without row exchange) are positive.

v. There is a rank n matrix R such that $A = R^T R$.

Proof.

• We saw $i. \Rightarrow ii.$ last section.

• For $ii. \Rightarrow i.$ observe that as A is symmetric, it is diagonalisable, so any vector x can be written in a basis of orthonormal eigenvectors

$$x = c_1 x_1 + c_2 x_2 + \dots c_n x_n$$

and so

$$\begin{aligned} x^T Ax &= x^T (c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 + \dots c_n \lambda_n x_n \\ &= c_1^2 \lambda_1 + c_2^2 \lambda_2 + \dots, c_n^2 \lambda_n > 0 \end{aligned}$$

• $ii. \Rightarrow iii.$: From i. it is easy to see that all upper left square submatrices also satisfy i. by considering vectors x ending in sufficiently many zeroes. So it is enough to observe that matrices satisfying ii. have positive determinants. But this is trivial as the determinant is the product of the eigenvalues.

• $iii. \Rightarrow iv.$ The pivots are the ratios of the determinants of consecutive upper left square submatrices.

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• *iv.* $\Rightarrow$ *v.* Since A is symmetric its (unique) LDU decomposition is the same as that of A^T, and so

$$LDU = A = A^T = U^T D^T L^T$$

implying that

$$A = LDL^T.$$

As the entries of D in the LDU decomposition are the pivots of A, we have that D can be written $D = \sqrt{D}\sqrt{D}$, and so

$$A = (L\sqrt{D})(\sqrt{D}L^T) = (\sqrt{(D)}^T L^T)^T(\sqrt{(D)}^T L^T) =: R^T R.$$

• *v.* $\Rightarrow$ *i.* Assuming $A = R^T R$ for R with rank n. We get

$$x^T Ax = x^T R^T Rx = (Rx)^T(Rx) = \|Rx\| \geq 0$$

As R has full rank, this is not 0 unless $x = 0$.

□

Now, a matrix A is *positive **semi**definite* if for all vectors x

$$x^T Ax \geq 0.$$

So for $A = \left[\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right]$ is positive semidefininte if $ac = b^2$, and the graph of $z = f_A(x, y)$ is a valley.

The principle submatrices are those you get by removing the same columns and rows. In this case is is not enough to consider only the upper left principle submatrices, as diagonals can be 0.

The characterisation theorem for positive definite matrics can be used to give a similar theorem from positive semidefinite matrices: one can essentially add small $\varepsilon > 0$ to the diagonal, apply the theorem for positive definite matrices, and then let ε go to 0. Doing this one gets:

Theorem 6.2. *For a symmetric real matrix A, the following are equivalent.*

i. $x^T Ax \geq 0$ *for all* x.

ii. All eigenvalues of A are non-negative.

iii. All principal submatrices of A have non-negative determinants.

iv. All pivots of A (without row exchange) are non-negative.

v. There is a matrix R (not necessarily rank n) such that $A = R^T R$.

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6.2.1 Positive definite matrics as Ellipses

The set of solutions of the polynomial equation $x^T Ix = 1$ or

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 1$$

is the surface of the unit n-sphere centered at the origin.

Changing co-efficients

$$d_1 x_1^2 + d_2 x_2^2 + \cdots + d_n x_n^2 = 1$$

shrinks the sphere in the x_i direction by a factor of $\sqrt{d_i}$. So for diagonal D $x^T Dx = 1$ is an ellipse if all entries of D are positive. (If some entries are 0 the ellipse 'becomes a cylinder' in that dimension. If they are negative we 'become hyperbolic' in that dimension.)

For a positive definite matrix A the diagonalisation

$$A = U\Lambda U^{-1}$$

by a unitary U (which doesn't change the lengths of vectors) yields that

$$1 = x^T Ax = x^T U\Lambda U^{-1} x$$

simply rotates the ellipse $x^T \Lambda x = 1$. Effectively, it is the ellipse we get from the unit sphere by stretching by a factor of $1/\sqrt{\lambda}$ in the direction of the ith eigenvector.

6.2.2 The Law of inertia

We look at a computational way to approximate (as closely as we like) the eigenvalues of a symmetric matrix.

Recall that for any matrix M, the matrix A shares eigenvalues with $B = M^{-1}AM$ (and has eigenvectors related through M). If M is orthogonal then B can be written $B = M^T AM$. We are going to try writing B like this for non-orthogonal M.

For symmetric A, and non-singular (but not necessarily orthogonal) C, we consider $C^T AC$, and call it *congruent* to A. As A is symmetric, so is $C^T AC$, and so its eigenvalues are still real. Indeed

$$(C^T AC)^T = (C^T A^T C)^T = (C^T AC).$$

But we have more than that its eigenvalues are real. The following is called Sylvester's Law of intertia.

Theorem 6.3. *Let A be a symmetric real matrix and C be a non-singular matrix. Then A and $C^T AC$ have the same number of positive, zero, and negative eigenvalues.*

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The proof is in the book. It is not diffifult, but uses notions that our not in our scope, so we omit it. Let's see why the theorem is useful though.

For symmetric A, we saw that the LDU factorisation is indeed a congruence

$$A = LDL^T$$

with a diagonal matrix D whose entries are the pivots of A. As these are the eigenvalues of D the above theorem gives us the following.

Corollary 6.4. *For symmetric A, the eigenvalues of A have the same signs as its pivots.*

This is nice, because finding pivots is easy, it is done by elimination. This allows us to approximate the eigenvalues of a symmetric matrix A as follows.

Example 6.5. Say a symmetric matrix A of dimension 10 has 3 positive pivots, so 3 positive eigenvalues. And say that $A - 7I$ has 1 positive eigenvalue.

As for any eigenvector x_i of A we have that

$$(A - 7I)x_i = \lambda_i x_i - 7x_i = (\lambda_i - 7)x_i,$$

the eigenvalues of $A - 7I$ are those of A shifted by 7. So we know that A has two eigenvalues between 0 and 7.

So we try with $A - \frac{7}{2}I$ etc. and with a lot of computation, can find one particular, or all, eigenvalues of A.

Problems from the text

6.2: 1, 2, 3, 8, 10, 12, 16, 19, 22 (2, 1, 5, 11, 13, 14, 21, 23, 20)

6.4 Minimum Principles

In Section 6.2 we saw that the set of critical points of $P_A(x) = x^T Ax$ for a symmetric matrix A was the solution space of $Ax = 0$. For the positive definite case, we could therefore have computed the unique critical point of $P_A(x)$ as $A^{-1}0$ had we not already known that it was $(0, 0)$. In this section we consider a more general quadratic polynomial than P_A and use these same ideas to find its critical point.

For a positive definite matrix A we started with the matrix equation $Ax = 0$, and multplied on the left by x^T to get the homgeneous quadratic polynomial

$$x^T Ax = x^T 0 = 0.$$

What happens when we multiply

$$Ax = b \qquad \text{and} \qquad Ax = \lambda x$$

by x^T.

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Ax = b

Multiplying $Ax = b$ or $Ax - b = 0$ by $x^T = (x_1, \ldots, x_n)$, we get

$$Q(x) := x^T Ax - x^T b = 0.$$

For a symmetric 2×2 matrix A this is

$$0 = Q(x) \quad = \quad \underbrace{x^2 a_{11} + 2a_{12}xy + y^2 a_{22}}_{P(X)} - b_1 x - b_2 y.$$

We can write $Q(x)$ as $Q(x) = P(x) - b_1 x - b_2 y$ so view it as a shift (by a plane) of the positve definite polynomial $P(X)$. Again, we find a critical point of $Q(x)$ with calculus. It occurs at

$$0 = Q_x = 2a_{11}x + 2a_{12}y - b_1 \tag{3}$$
$$0 = Q_y = 2a_{12}x + 2a_{22}y - b_2 \tag{4}$$

Which is where

$$0 = 2A \begin{bmatrix} x \\ y \end{bmatrix} - b \qquad \text{so} \qquad 2Ax = b.$$

This is at $x = \frac{1}{2}A^{-1}b$.

Note

We find the critical point of $Q(x)$ by solving $Ax = b$. So where is this $\frac{1}{2}$ term coming from? Well, treat A and b as constants and x simply as a single variable. Then $Q(x)$ becomes $Q(x) = Ax^2 - bx$. To find the critical points of this we solve $0 = Q'(x) = 2Ax - b$ and get $x = \frac{1}{2}\frac{b}{A}$. So maybe the 1/2 is to be expected.

Multiplying A by 1/2 multiplies A^{-1} by 2 which gives a more convenient form for what we have shown:

$$Q(x) = \frac{1}{2}x^T Ax - x^T b + c$$

has a critical point at $x_* = A^{-1}b$. Putting this into $Q(x)$ gives a formula for the minimum value of $Q(x)$.

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$$
\begin{aligned}
Q(x_*) &= \frac{1}{2}x_*^T A x_* - x_*^T b + c \\
&= \frac{1}{2}(A^{-1}b)^T A (A^{-1}b) - (A^{-1}b)^T b + c \\
&= \frac{1}{2}b^T (A^{-1})^T b - b^T (A^{-1})^T b + c \\
&= -\frac{1}{2}b^T (A^{-1})^T b + c \\
&= -\frac{1}{2}b^T A^{-1} b + c \\
&= -\frac{1}{2}b^T x_* + c
\end{aligned}
$$

We can use this to find the minimim and minimum value of a quadratic polynomial.

Example 6.6. Minimize $Q(x) = 2x_1^2 - x_1x_2 + 3x_2^2 - x_1 + x_2$.

We can write this as $Q(x) = \frac{1}{2}x^T Ax - x^T b$ where

$$A = \begin{bmatrix} 4 & -1 \\ -1 & 6 \end{bmatrix} \text{ and } b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

So its minimum occurs as

$$x_* = A^{-1}b = \frac{1}{23}\begin{bmatrix} 6 & 1 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{23}\begin{bmatrix} 5 \\ -3 \end{bmatrix}$$

and is

$$
\begin{aligned}
Q(x_*) &= -\frac{1}{2}(b^T x_*) + c \\
&= -\frac{1}{2}(\begin{bmatrix} 1 & -1 \end{bmatrix}\frac{1}{23}\begin{bmatrix} 5 \\ -3 \end{bmatrix}) = \frac{-8}{46}
\end{aligned}
$$

Practice

Find the critical point of

$$Q(x) = 2x_1^2 + 3x_2^2 + x_3^2 - x_1x_2 + x_1x_3 + 2x_2x_3 + x_1 - x_2 + x_3 + 5$$

using the same method as above. Check that it agrees with what you would get using just calculus.

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An application: Minimising with Constraints

What is the min of $Q(x) = x^T Ax - x^T b$ when constrained to

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} x = c_1 x_1 + c_2 x_2 = d?$$

Legranges method writes

$$L(x, y) = Q(x) + y(c_1 x_1 + c_2 x_2 - d)$$

and then uses the fact that at for a critical point (x, y) we have

$$0 = L_y = c_1 x_1 + c_2 x_2 - d.$$

So our constraint is satisfied at critcal points of $L(x, y)$. Moreover, as this is satisfied, we have at critical points that

$$L(x, y) = Q(x) + y(0) = Q(x).$$

Thus any local minimum $L(x, y)$ is a local minimum of $Q(x)$ when restricted to the constraint.

Where x, b and A had dimension n, we augment them with the constraint, writing

$$x' = \begin{bmatrix} x_1 \\ \vdots \\ x_n \\ y \end{bmatrix} \qquad A' = \left[\begin{array}{ccc|c} & & & c_1 \\ & A & & \vdots \\ & & & c_n \\ \hline c_1 & \cdots & c_n & 0 \end{array}\right] \qquad b' = \begin{bmatrix} b_1 \\ \vdots \\ b_n \\ d \end{bmatrix}$$

and so get

$$L(x, y) = \frac{1}{2} x'^T A' x' - x'^T b'.$$

Our critical point is at $x'_* = A'^{-1} b'$ and has value $L(x'_*) = -\frac{1}{2} b'^T x'_*$.

Practice

Where $Q(x)$ is as in Example 6.6 find its minimum value constrained to

$$2x_1 - x_2 = 4.$$

The Rayleigh Quotient

The *Rayleigh quotient* of an n-dimensional vector x with respect to an $n \times n$ matrix A is

$$R(x) = R_A(x) := \frac{x^T Ax}{x^T x}.$$

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When x is an eigenvalue of A this becomes

$$R(x) = \frac{x^T Ax}{x^T x} = \frac{x^T \lambda x}{x^T x} = \lambda = \pm \frac{\|Ax\|}{\|x\|}.$$

So for eigenvectors x of A the Rayleigh quotient $R_A(x)$ is a measure of how much transformation by A stretches x. This intuition goes beyond eigenvectors, as in general we have that

$$\frac{\|Ax\|^2}{\|x\|^2} = \frac{(Ax)^T Ax}{x^T x} = R_{A^T A}(x). \tag{5}$$

Note

For investigation of the Rayleigh quotient, observe that for any constant c we have

$$R(cx) = \frac{(cx)^T A(cx)}{(cx)^T (cx)} = \frac{c^2 x^T Ax}{c^2 x^T x} = R(x)$$

and so we can see all there is to see about it by considering normalized vectors x for which we have the simpler formula $R(x) = x^T Ax$.

For these bits we are using the convention that the eigenvalues (which are all real) are ordered with $\lambda_i \leq \lambda_{i+1}$.

When A is symmetric, we diagonalise it as $A = Q\Lambda Q^T$ by unitary Q and get, where $x = Qy$ for some y which also has unit length, that

$$R(x) = x^T Ax = (Qy)^T A(Qy) = y^T \Lambda y = \lambda_1 \|y_1\|^2 + \cdots + \lambda_n \|y_n\|^2$$

from which it is clear that $R(x)$ is largest when $x = x_n$ and smallest when $x = x_1$. That is

$$\lambda_1 \leq R(x) \leq \lambda_n.$$

Moreover as $A^T A = A^2$ and the eigenvalues of A^2 are the squares of those of A we get from this and (5), by taking square roots, that

$$|\lambda_1| \leq \frac{\|Ax\|}{\|x\|} \leq |\lambda_2|.$$

Another useful observation is that

$$R(e_i) = \frac{e_i^T Ae_i}{e_i^T e_i} = a_{ii}$$

so we have the following bounds on the eigenvalues that are useful if we want to approximate them computationally:

$$\lambda_1 \leq a_{ii} \leq \lambda_n.$$

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Observe also that for the matrix B which we get from A by removing the ith row and column, we get that $x^T Bx = 1$ is the $n-1$ dimensional ellipsoid which is the intersection of $x^T Ax = 1$ with a hyperplane. So the princible axis of $x^T Bx = 1$ is at most that of $x^T Ax = 1$, yielding that $\lambda_{n-1}(B) \leq \lambda_n(A)$. Extending this argument we get the *interlacing of eigenvalues*

$$\lambda_1(A) \leq \lambda_1(B) \leq \lambda_2(A) \leq \lambda_2(B) \leq \cdots \leq \lambda_{n-1}(B) \leq \lambda_n(A).$$

Problems from the text

6.4: 1, 2, 3, 9, 11, 14 (1, 3, 5, 10, 8, 16)

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7 Computations with matrices

We can solve $Ax = b$ for an $n \times n$ matrix A well enough algebraically in $O(n^3)$ basic operations by Gaussian Elimination. There are reasons we might prefer to find approximate solutions. One is that we can can, at least for large matrices, reduce the number of basic operations. The other reason is that this term 'basic operations' that we have been using is a little deceptive. When we say we can solve $Ax = b$ in $O(n^3)$ basic operations, we mean $O(n^3)$ multiplications, divisions or additions of elements of $\mathbb{R}$. When we are computing a solution algebraically, even if we start with single digit integers, these entries grow through the process of Gaussian Elimination. When you compute the inverse of a 3×3 matrix of single digit integers, you often get an entry expressed as a fraction of two digit numbers. Something like: $\frac{11}{3}$. For a 7×7 matrix it is not difficult to get an entry such as $\frac{-4842445}{23114472}$ in the inverse. It is not much bigger than the original entries, but it requires more digits to represent, and basic operations for such numbers take more time. So we soon throw away the exact solution, and keep only a decimal approximation

$$\frac{-4842445}{23114472} \approx -0.20949.$$

Generally when doing comptations on bigger matrices, we save a LOT of time and space by rounding numbers to some fixed precision. We should known how much this rounding effects our result.

We look at this in this chapter. The first step is to introduce a 'measure of goodness' for approximations.

7.2 Matrix Norm and Condition Number

When solving $Ax = b$ we want to known when it is okay to approximate A or b. That is, we want to know that if we change A or b by some small bit we will change the solution $x = A^{-1}b$ by only some small bit.

Let's consider first changing only b. If we solve $Ax = b$, by first replacing b with

$$b' = b + \varepsilon_b$$

for some small error ε_b. Our solution will be $A^{-1}(b + \varepsilon_b) = x + A^{-1}\varepsilon_b$ which adds to x some small error $\varepsilon_x = A^{-1}\varepsilon_b$. We want to measure ε_x compared to ε_b. So that we can say a $\%p$ change in b yields a $\%p'$ changed in x, we normalize the errors $\varepsilon_b = A\varepsilon_x$ and ε_x with respect to $b = Ax$ and x.

The *condition number* of A is the maximum c such that

$$\frac{\|\varepsilon_x\|}{\|x\|} < c\frac{\|A\varepsilon_x\|}{\|Ax\|}, \tag{6}$$

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over all x and ε_x. Equivalently we can write this as

$$\frac{\|Ax\|}{\|x\|} < c\frac{\|A\varepsilon_x\|}{\|\varepsilon_x\|}, \tag{7}$$

and so interpret the condition number as the (maximum A stretches a vector) over the (minimum A stretches a vector).

If A has a small condition number, then a small relative change in b results in a small relative change in x. So, given A, how do we compute this c?

7.2.1 Condition number of a positive definite A

Considering the case that A is posive definite, the min and max eigenvalue λ_m and λ_M are positive integers, and we get:

$$\lambda_m \leq \frac{\|Ax\|}{\|x\|} \leq \lambda_M.$$

So as $Ax = b$ and $A\varepsilon_x = \varepsilon_b$ we have

$$\frac{\|b\|}{\|x\|} \leq \lambda_M \qquad \text{and} \qquad \lambda_m \leq \frac{\|\varepsilon_b\|}{\|\varepsilon_x\|}.$$

Plugging these into (7) yields $c \leq \lambda_M/\lambda_m$, and taking x as an eigenvector for λ_M and ε_x as one for λ_m we get equality, so

$$c = \frac{\lambda_M}{\lambda_m}$$

is the condition number for A.

Practice

If A is orthogonal, $b = (1, 3, 4, 1)$ and $\varepsilon_b = (\frac{1}{10}, 0, \frac{2}{10}, \frac{1}{10})$, and $Ax = b$ and $A\varepsilon_x = \varepsilon_b$, then what is (an upper bound for) $\|\varepsilon_x\|\|x\|$?

Practice

Do A and A^{-1} have the same condition number?

7.2.2 Condition number of more general A

When A is non-symmetric (non-positive definite) what are the maximum and minimum values of

$$\frac{\|Ax\|}{\|x\|}?$$

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Still the max the min eigenvalues?

No! The proof on this bound used ellipses in the symmetric case. It required that there was a full complement of eigenvectors.

Example 7.1. The matrix $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ has eigenvalues $\lambda_1 = \lambda_2 = 1$ with a single eigenvector $(1, 0)$. For a vector $v = (2, 2)$ we have

$$Av = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

So $\|Av\|/\|v\| = \frac{\sqrt{20}}{\sqrt{8}} = \sqrt{5/2} > 1$

Definition 7.2. The *norm* of a matrix A is

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}.$$

This immediately yields the useful identity

$$\frac{\|Ax\|}{\|x\|} \leq \|A\| \text{ or } \|Ax\| \leq \|A\|\|x\|. \tag{8}$$

Practice

Show that if A is invertible, then

$$\|A^{-1}\| = \max_{y \neq 0} \frac{\|y\|}{\|Ay\|}.$$

As the condition number c of A is the maximum value of $\frac{\|Ax\|}{\|x\|} / \frac{\|Ay\|}{\|y\|}$ we get it by maximising the top of the fraction, and minimizing the bottom, so get $c = \|A\|\|A^{-1}\|$.

7.2.3 Error in A

Until now, we approximated $Ax = b$ by altering b. Commonly we will round all our numbers, and introduce error in A as well. Solving $(A + \varepsilon_A)(x + \varepsilon_x) = b$ we get

$$b = Ax + \varepsilon_A x + A\varepsilon_x + \varepsilon_A \varepsilon_x.$$

As we know that $Ax = b$ we therefore get that $A\varepsilon_x + \varepsilon_A(x + \varepsilon_x) = 0$ and rearranging this gives that

$$-\varepsilon_x = A^{-1}\varepsilon_A(x + \varepsilon_x).$$

Using (8) twice we get

$$\|\varepsilon_x\| \leq \|A^{-1}\| \cdot \|\varepsilon_A(x + \varepsilon_x)\| \leq \|A^{-1}\| \cdot \|\varepsilon_A\| \cdot \|(x + \varepsilon_x)\|,$$

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which yields

$$\frac{\|\varepsilon_x\|}{\|(x+\varepsilon_x)\|} \leq \|A^{-1}\|\|\varepsilon_A\| = c\frac{\|\varepsilon_A\|}{\|A\|}.$$

So the relative error in x is again bound by the condition number of A times the relative error in A.

We want to compute norm then.

Well,

$$\|A\|^2 = \max \frac{\|Ax\|^2}{\|x\|^2} = \max \frac{x^T A^T Ax}{x^T x} = \max R(x)$$

where $R(x)$ is the Rayleigh Quotient of x with respect to $A^T A$. We saw that the Rayleigh quotient is maximized at the maximum eigenvector. So $\|A\|$ is the root of the max eigenvector of $A^T A$.

Problems from the text

7.2: 2, 5, 9, 10, 12, 15, 17 (4, 1, 5, 7, 14, 12, 13)

7.3 Computing Eigenvalues

There are many techniques for computing eigenvalues, and we will see two basic ones in this section: the power method and the QR-method. The power method is intuitive– it is very easy to see why it works and to see why such simple improvements as 'shifting' work. The QR-method is less transparent, but faster; it is called the QR-method because it uses a QR-decomposition of a matrix A. There are various ways to find a QR-decomposition of a matrix. We have seen one, the Gram-Schmidt method. We will see another using so-called Householder matrices.

Not only the power method and the QR-method, but other methods for computing eigenvalues are much faster for matrices with many zero entries. So the first step in computing the eigenvalues of A is usually to find a similar matrix with lots of zero entries. Finding the Jordan form of the matrix would be ideal, but we don't have that if we don't have the eigenvalues. We will see a method of putting a matrix into a Hessenberg form, using the same Householder matrix.

Though the algorithm for the Hessenberg form would be used before finding its QR-decomposition, we present the QR-decomposition via Householder matrics first as it is easier to understand than the Hessenberg algorithm, and helps to explain why the Hessenberg algorithm works.

7.3.1 The Householder Matrix

Recall that the projection matrix P_v such that for any vector x, $P_v x$ is the projection of x onto v is computed as $P_v = \frac{vv^T}{v^T v}$.

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For a vector $x \in \mathbb{R}^n$, let z be the first unit normal vector e_1 and $\sigma = \|x\|$, and let

$$H = H_x := I - 2P_{(x+\sigma z)} = I - 2\frac{(x+\sigma z)(x+\sigma z)^T}{\|x+\sigma z\|^2}.$$

Lemma 7.3. *Let $H = H_x$ be the Householder matrix for a vector x. The following are true.*

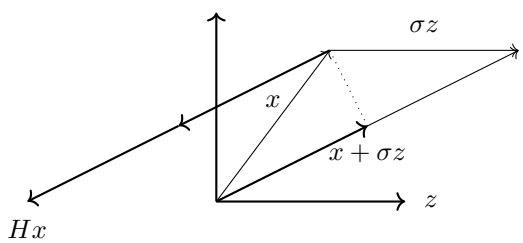
i. $H_x x = -\sigma z$

ii. *H is invertible*

iii. *H is symmetric*

iv. *H is orthogonal*

Proof. We prove *i.* with a picture:



As the only vector H takes to 0 is 0, it is invertible, which gives *ii.*

To see *iii.*, that H is symmetric, observe that on normalising $x + \sigma z$ to u H can be written as $H = I - 2uu^T$. So

$$H^T = (I - 2uu^T)^T = I - 2(u^T)^T u^T = I - 2uu^T = H.$$

To see *iv.*, that H is orthogonal, observe that

$$H^T H = H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4u\overbrace{u^T u}^{=1}u^T = I^2 = I.$$

□

7.3.2 QR-decomposition with the Householder matrix

Recall the QR-decomposition of a square matrix A was a decomposition $A = QR$ into a orthogonal matrix Q and an upper triangular matrix R. We found this for invertible A by using the Gram-Schmidt process on the columns $b_1, \ldots, b_n$

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of A to get the equivalent orthonormal basis $Q = \{q_1, \ldots, q_n\}$ of the column space. Gram-Schmidt assured us that when writing b_i in the basis Q the vector

$$b_j = r_{1,j}q_1 + \ldots r_{n,j}q_n = (q_1^T b_j)q_1 + \cdots + (q_n^T b_j)q_n$$

had $r_{i,j} = 0$ if $I > j$. Thus letting Q be the matrix with vectors of Q as its columns and $R = [r_{ij}]$, we get that $A = QR$ is a QR-decomposition of A.

For two square matrices A and B the matrics $A \oplus B$ is the block diagonal matrix with A in the upper left corner and B in the lower left corner. If n is fixed and A is a matrix of dimension $n - i$ for some i, when we write $I \oplus A$ it is understood that I is the i dimensional identity, so that $I \oplus A$ is an $n \times n$ matrix.

We now get a QR-decompostion of A using Householder matrices. Let $b_1, b_2 \ldots, b_n$ be the columns of $A =: A_0$. Where $U_1 = H_1 = H_{b_1}$ is the Householder matrix for b_1, we get that

$$A_1 := U_1 A = \left[\begin{array}{c|c} -\sigma_1 & ? \\ \hline 0 & \\ 0 & ? \\ 0 & \end{array}\right]$$

has zeros under the diagonal of the first column.

Now we would like to clear under the diagonal of the second column. Multiplying on the right by H_{b_2} (where we used the second standard normal vector for z rather than the first) would do this, but it would mess up our nice first column at the same time. To avoid this we want something resembling H_{b_2} but which has zeros under the diagonal of its first column.

Let A_1^* be the matrix we get from A_1 by removing the first row and column. Let H_2 be the householder matrix H_{b_2*} of the first column of A_1^*, and let $U_2 = I \oplus H_2$.

Then multiplying A_1 by U_2 clears under the diagonal in the second column.

$$U_2 U_1 A = \left[\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & & & \\ 0 & & H_2 & \\ 0 & & & \end{array}\right] \left[\begin{array}{c|c} -\sigma_1 & ? \\ \hline 0 & \\ 0 & ? \\ 0 & \end{array}\right] = \left[\begin{array}{cc|c} -\sigma_1 & ? & ? \\ 0 & ? & \\ \hline 0 & & ? \\ & & \end{array}\right].$$

As H_2 is symmetric and orthogonal, so is U_2. So U_2U_1 is orthogonal too. Proceeding recursively, we get the following.

The Householder QR-decomposition of A.

To find the QR decomposition of A, let $A_0 = A$. For $i = 1$ to n let

- A_i^* be the lower right $(n-i) \times (n-i)$ submatrix of A_i,

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- H_i be the householder matrix of its first column, and
- $U_i = I \oplus H_i$.

Then $Q^T = U_n U_{n-1} \dots U_1$ is orthogonal and $R = Q^T A$ is upper-triangular.

Practice

Justify the following claims we made above:

i. If H_i is orthogonal then so is $U_i = I \oplus H_i$.

ii. If U_i and U_j are orthognal, then so is $U_i U_J$.

Practice

QR-decompositions of A are not unique, as multiplying a column q_i of Q by -1 does not change the fact that the columns are orthanormal, and this gives another QR-decomposition in which the entries $r_{i,j}$ have been multiplied by -1.

i. Show that if one insists that the diagonals of R are positive, then there is a unique QR-decomposition of A.

ii. Argue that in the QR-decomposition one gets by the Gram-Schmidt process the diagonal entries of R are all positive.

iii. Conclude that if $A = QR$ is the decomposition we got above, then $A = (-Q)(-R)$ is the Gram-Schmidt QR-decomposition.

This gives us another way to compute the QR-decomposition of a matrix A, and R has lots of nice zeros. Unfortunatley, it is not similar to A, so this doesn't help us in our task of finding a matrix similar to A with lots of zeros. What if we multiply the $U_i = U_i^{-1}$ on the right of A as well. This would give us something similar to A, but the first column of

$$U_1 A U_1 = H_1 A H_1 = \left[\begin{array}{c|c} \sigma_1 & ? \\ \hline 0 & \\ 0 & ? \\ 0 & \end{array}\right] H_1$$

is no longer mostly zeros. The first column would remain unchanged if we multiplied on the right by $\left[\begin{smallmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & H_1' & \\ 0 & & & \end{smallmatrix}\right]$ for some H_1' instead of H_1. Nice. Oh, but shoot, then we have to do the same on the left too. It turns out that works okay.

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7.3.3 Making A almost upper triangular (Hessenberg Form)

Decompose $A_0 := A$ as follows.

$$A_0 = \left[\begin{array}{c|c} a_{11} & ? \\ \hline a_1^* & A_0^* \end{array}\right]$$

With this, we see that

$$A_1 := \underbrace{\left[\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & & & \\ 0 & & H_1 & \\ 0 & & & \end{array}\right]}_{U_1^{-1}} \left[\begin{array}{c|c} a_{11} & ? \\ \hline a_1^* & A_0^* \end{array}\right] \underbrace{\left[\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & & & \\ 0 & & H_1 & \\ 0 & & & \end{array}\right]}_{U_1} = \left[\begin{array}{c|c} a_{11} & ? \\ \hline H_1 a_1^* & ? \end{array}\right]$$

So taking H_1 as the householder matrix for a_1^*, this becomes

$$\left[\begin{array}{c|c} a_{11} & ? \\ \hline ? & \\ 0 & ? \\ 0 & \end{array}\right]$$

in which the first column is 0 under the subdiagonal. We now do the same to the second column.

Our A_1 decomposes as

$$U_1 A U_1 = \left[\begin{array}{cc|c} ? & ? & ? \\ ? & ? & ? \\ \hline 0 & a_2^* & A_2^* \\ 0 & & \end{array}\right].$$

Letting H_2 be the householder matrix for a_2^* and $U_2 = I \oplus H_2$, conjugating A_1 by U_2: $A_2 = U_2 A_1 U_2$, clears under the subdiagonal in the second column. We continue in this fashion.

The Hessenburg Form of A

To find the Hessenburg Form of A, let $A_0 = A$. For $i = 1$ to n let

- A_{i-1}^* be the lower right $(n-i) \times (n-i)$ submatrix of A_{i-1},
- a_i^* be the ith column of A_{i-1} with its first i entries removed,
- H_i be the householder matrix of a_i^*, and
- $U_i = I \oplus H_i$.

Then $Q = U_n U_{n-1} \dots U_1$ is orthogonal and $H = QAQ = QAQ^{-1}$ has all zeros under the subdiagonal. If A is symmetric then so is H, and so it is tridiagonal.

Example 7.4. We tridiagonalise $A = \left[\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{smallmatrix}\right]$. Writing it as $\left[\begin{array}{c|cc} 2 & 1 & 1 \\ \hline 1 & 1 & 2 \\ 1 & 2 & 3 \end{array}\right]$, our a_1^* is $x = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ so H_1 is the householder matrix $I - 2P_{x+\sqrt{2}z}$ is

$$I - 2(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \sqrt{2}\begin{bmatrix} 1 \\ 0 \end{bmatrix})(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \sqrt{2}\begin{bmatrix} 1 \\ 0 \end{bmatrix})^T.$$

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We (use a computer to) get

$$H_1 = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} - \frac{1}{1+\sqrt{2}} \begin{bmatrix} 3+2\sqrt{2} & 1+\sqrt{2} \\ 1+\sqrt{2} & 1 \end{bmatrix} = \frac{\sqrt{2}}{2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}.$$

This gives our Hessenburg from of A:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} 2 & -\sqrt{2} & 0 \\ -\sqrt{2} & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}.$$

Now for any matrix A, we can get a similar matrix almost half of the entries of which are 0. Using this will reduce the work in iterative methods for computing the eigenvalues of A.

Note:

Recall that when $a_1, a_2, a_3, \ldots$ is a series of numbers limiting to a value a, the *rate of convergence* of the sequence is

$$\lim_{n\to\infty} \frac{|a - a_{n+1}|}{|a - a_n|}.$$

If this is less than 1 then the series converges to a. A small rate (close to 0) is better. It means that the sequence converges more quickly. If

$$\lim_{n\to\infty} \frac{|a - a_{n+1}|}{|a - a_n|^2} = c < 1,$$

then the convergence is *quadratic*. This is even better.

7.3.4 The Power Method

If your maximum eigenvalue is complex, for example, then the conjugate will have the same absolute value and so the power method doesn't work so well. But if A is symmetric, then this is not a problem, and most random symmetric matrices will be okay.

For this section we order the eigenvalues of a matrix A such that $|\lambda_1| \leq |\lambda_2| \leq \cdots \leq |\lambda_n|$, and call λ_1 the minimum eigenvalue and λ_n the maximum.

This method works to find the maximum eigenvalue λ_n of A as long as $|\lambda_{n-1}| \neq |\lambda_n|$. Indeed, any vector is of the form

$$u = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$$

and almost any random vector has $c_n \neq 0$.

If this holds, then

$$A^k u = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \cdots + c_n \lambda_n^k x_n.$$

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Normalising this to

$$u_k = \frac{A^k u}{\|A^k u\|} = \frac{1}{\|A^k u\|} c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \cdots + c_n \lambda_n^k x_n$$

we get that u_k approaches $\frac{x_n}{\|x_n\|}$ which is a unit eigenvector for λ_n. Once we have this, we can compute $\|\lambda_n\| = \frac{\|Ax_n\|}{\|x_n\|}$. Inspecting the ratios indivual entries of these two vectors, they should all (or most) have the same sign, telling us the sign of λ.

Note

The rate of convergence is $\frac{|\lambda_{n-1}|}{|\lambda_n|}$, so this converges better when there is a big gap between λ_n and the next eigenvalue.

In the next several examples we will 'find' the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 0 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. For reference, they are approximately

$$\begin{aligned} \lambda_1 &\approx 1.913 & x_1 &\approx (0.751, 0.188, -0.631) \\ \lambda_2 &\approx -2.751 & x_2 &\approx (0.462, -0.833, 0.301) \\ \lambda_3 &\approx 6.838 & x_3 &\approx (0.469, 0.519, 0.714) \end{aligned}$$

Example 7.5. We find the maximum eigenvalue λ_3 of the matrix $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 0 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, by choosing a random vector $u = (3, 4, 5)$ and computing $u_i = A^i u$ for several i. Our normalised iterations, approximations of x_3, are

$$\frac{u_1}{\|u_1\|} = (.476, .497, .725), \frac{u_2}{\|u_2\|} = (.463, .527, .712), \frac{u_3}{\|u_3\|} = (.471, .515, .715), \ldots$$

This seems close. To compute the maximum eigenvalue λ_3 we then we compute

$$|\lambda_3| \approx \frac{\|Au_3\|}{\|u_3\|} \approx 6.838.$$

Comparing $u_3 = (1063, 1164, 1615)$ and $Au_3 = (7233, 8034, 11015)$ we conclude that λ is positive, so is about 6.838.

We expected a rate of convergence of about .4 here. We started with a pretty good guess, and this should give us another digit of accuracy every 2 or 3 steps, so upto three decimal points we should be pretty good after $8 - 10$ iterations.

Now, how can we get the next eigenvalue, λ_2? If we remove the projection of u_0 onto x_3, then when we mutliply by A, it is the x_2 component that grows the fastest.

Example 7.6. Letting $u_0 = (3, 4, 5) - \text{proj}_{x_3}(3, 4, 5) = (-0.669, 0.738, -0.077)$ we compute

$$u_1 = \frac{Au_0}{\|Au_0\|} = (0.298, -0.835, 0.461).$$

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Now if we had really used x_3 in computing u_0 we could just iterate as above, but have used an approximation of x_3, and we have rounded our result. So there is still some non-zero projection of u_1 onto x_3. We should remove this at each iteration.

$$u_2 = \frac{A(u_1 - \operatorname{proj}_{x_3} u_1)}{\|A(u_1 - \operatorname{proj}_{x_3} u_1)\|} = (-0.576, 0.793, -0.197)$$
$$u_3 = \frac{A(u_2 - \operatorname{proj}_{x_3} u_2)}{\|A(u_2 - \operatorname{proj}_{x_3} u_2)\|} = (0.376, -0.849, 0.369)$$

In a couple more approximations we should be able to compute λ_2. Notice, we see already that it is negative, our signs are all flipping back and forth.

If we wanted, we could continue on and compute x_1 with a guess $u_0 = a - \operatorname{proj}_{x_2} a - \operatorname{proj}_{x_3} a$, in the same way.

Also notice that we could compute x_3 and x_2 as in the previous examples at the same time. Start with initial u_0, which we use to estimate x_3 and v_0 which use to estimate x_2. We compute $\begin{bmatrix} u_1 \mid v_1 \end{bmatrix} = A \begin{bmatrix} u_0 \mid v_0 \end{bmatrix}$. Then apply Gram Schmidt, replacing v_1 with $v_1 - \operatorname{proj}_{u_1} v_1$, and normalising both columns. As u_1 is approximating x_3 doing this is approximately removing the x_3 component from v_1. Iterate this, using Gram Schmidt at each step.

7.3.5 The Inverse Power Method

Doing the same for A^{-1} we can find λ_1. The rate of convergence of the inverse power method is $|\lambda_1|/|\lambda_2|$.

Example 7.7. Let's find the minimum eigenvalue λ_1 of $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 0 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. From a first guess $u_0 = (3, 4, 5)$ we compute $u_i = (A^{-1})^i u_0$ and normalising get approximations of x_1:

$$\frac{u_1}{\|u_1\|} = (.477, .371, .796), \frac{u_2}{\|u_2\|} = (.083, .713, .695), \frac{u_3}{\|u_3\|} = (.114, -.269, .956), \ldots$$

This doesn't seem so good. We started with a bad guess, and we expect convergence to be slower, as $|\lambda_1|/|\lambda_2| \approx 0.694$. But let's use it to estimate the eigenvalue:

$$|\lambda_1'| \approx \frac{\|Au_3\|}{\|u_3\|} \approx 0.371.$$

This is the eigenvalue for A^{-1}. Our estimate at λ_1 is 2.688.

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There is a nice way that we can boost the rate of convergence considerably when we start with a good approximation σ of λ_1. This is called the *shifted inverse power method*. We simply do the inverse power method on $A - I\sigma$. This has eigenvalues $\lambda_1 - \sigma, \lambda_2 - \sigma, \ldots, \lambda_n - \sigma$, so the inverse power method converges with rate $|\lambda_1 - \sigma|/|\lambda_2 - \sigma|$.

Example 7.8. Letting $B = A - 2.688I$ (alternately we could use $A^{-1} - .371I$) we apply inverse power method to B. Computing $u_i = B^{-i}u_0$ with the same initial guess $u_0 = (3, 4, 5)$ we get

$$\frac{u_1}{\|u_1\|} = (0.571, 0.497, 0.652) \qquad \frac{u_2}{\|u_2\|} = (0.002, 0.374, 0.927)$$
$$\frac{u_3}{\|u_3\|} = (0.861, 0.327, -0.387) \qquad \frac{u_4}{\|u_4\|} = (-0.723, -0.157, 0.671)$$
$$\frac{u_5}{\|u_5\|} = (0.757, 0.194, -0.623) \qquad \frac{u_6}{\|u_6\|} = (-0.751, -0.187, 0.633)$$
$$\frac{u_7}{\|u_7\|} = (0.752, 0.188, -0.631)$$

The eigenvalues for B are $\{-2.751, 1.913, 6.838\} - 2.681 = \{-5.432, -.768, 4.157\}$, so the rate of convergence is about $\frac{.768}{4.15} = .18$. Even starting with a back first guess as we did, convergence was pretty good.

Note that if our estimate at λ_1 is too big, we can overshift, and the min eigenvalue of $A - sI$ might not be the one we get from that of the min eigenvalue of A. If λ_2 of A had been positive this would have happened here. Perhaps one halves the shift to be conservative to avoid this. Or perhaps one is just happy finding x_2 and λ_2 instead. How would you know that you have done this?

All of these versions of the power method have their uses, but tend to be slow. In practice the QR-method works better, though it is less intuitive.

7.3.6 The QR-method

We can compute the QR-decomposition of A using Gram-Schmidt or Householder matrices or in the text book there is another method based on subspace rotation that is even faster. However we do it, we can use this decomposition to iteratively triangulate A via the QR-method.

We can decompose $A_0 = A = QR$ where Q is orthogonal and R is upper triangular, so $R = Q^{-1}A$. The matrix $A_1 = RQ = Q^{-1}RQ$ is similar to A_0 and as it is the product of some Q with an upper triangular R, it tends to have most of its weigth up in the upper right traingle. This is a vague statement, that we will not prove, but lets see what we mean with an example.

Example 7.9. The matrix $A_0 = A = \begin{bmatrix} -.66 & .58 & -.52 \\ .68 & -.20 & .47 \\ .26 & -.81 & -.42 \end{bmatrix}$ has (approximate) QR

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decomposition $A_0 = Q_0 R_0$ where

$$Q_0 = \begin{bmatrix} .69 & -.69 & -.28 \\ -.11 & -.44 & .88 \\ -.75 & -.56 & -.44 \end{bmatrix} \text{ and } R_0 = \begin{bmatrix} 1 & .75 & -.56 \\ 0 & -.75 & -.56 \\ 0 & 0 & .28 \end{bmatrix}.$$

This gives $A_1 = R_0 Q_0 = \begin{bmatrix} -.38 & .62 & 1.2 \\ & .50 & .62 & -.41 \\ -.20 & -.16 & -.10 \end{bmatrix}$. Taking the QR factorisation $A_1 = Q_1 R_1$ of this we can then compute $A_2 = R_1 Q_1 = \begin{bmatrix} .5 & .31 & -1 \\ .-06 & .56 & .75 \\ 0 & -.19 & -.5 \end{bmatrix}$.

The entries in the top right are getting larger (in absolute value) while those in the lower left are getting smaller. Continuing in this way, we get a series of matrices similar to A that are getting closer and closer to upper triangular, so the diagonals are approaching the eigenvalues of A.

Problems from the text

7.3: 1, 2, 3, 6 ('same A' refers to quesion 1), 7, 8, 10 (1, 3, 5, 2, 8, 6, 7)

7.4 Iterative Methods for solving $Ax = b$

We discussed at the beginning of the chapter why we will often use approximate values in solving a matrix equation such as $Ax = b$ rather than finding the precise algebraic solution. We argued that in computing a algebraic solution, the 'basic operations' became more expensive. But even ignoring this, iterative methods can give an improvement over Gaussian Elimination in the number of basic operations. Recall, Gaussian Elimination takes $O(n^3)$ operations, and back-substitution $O(n^2)$. With iterative methods we can replace elimination with calculations that take about n^2 operations. For big matrices this is important.

The techniques we look at for solving $Ax = b$ all involve splitting the matrix A as $A = S - T$ for some invertible S. If we do this we have

$$\begin{aligned} Ax = b \quad &\iff \quad (S - T)x = b \\ &\iff \quad Sx - Tx = b \\ &\iff \quad Sx = Tx + b \\ &\iff \quad x = S^{-1}Tx + S^{-1}b \end{aligned}$$

Starting with an approximation x_1 of x we then calculate

$$x_2 = S^{-1}Tx_1 + S^{-1}b.$$

So why do we expect x_2 to be a better approximation of x? Well, let $\varepsilon_1 =$

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$x - x_1$ be the error in our first approximation. We have

$$\begin{aligned} x &= S^{-1}Tx + S^{-1}b \\ &= S^{-1}T(x_1 + \varepsilon_1) + s^{-1}b \\ &= S^{-1}Tx_1 + S^{-1}b + S^{-1}T\varepsilon_1 \\ &= x_2 + S^{-1}T\varepsilon_1 \end{aligned}$$

As long as $S^{-1}T\varepsilon_1 < \varepsilon_1$, x_2 is closer to x than x_1 is. We can assure this if $S^{-1}T$ has norm less than 1. So the trick is to find S and T such that $\|S^{-1}T\| < 1$ is small, and such that S^{-1}, $S^{-1}b$ and $S^{-1}T$ are quick to calculate. Once we have these, computing $x^2 = S^{-1}Tx_1 + S^{-1}b$ takes about $n^2 + n$ operations, so if we iterate only constantly many times, we save time for large enough n.

7.4.1 Jacobi's Method

Jacobi's method takes S as the diagonal of A. As S is diagonal, computing S^{-1} is a quick n operations. Computing $S^{-1}b$ and $S^{-1}T$ take n and n^2 operations, so we have $n^2 + 2n$ one time operations. Computing $x_{i+1} = S^{-1}Tx_i + S^{-1}b$ takes $n^2 + n$ each time we iterate.

Example 7.10. We use Jacobi's method to approximate the solution $x \approx (-1.2, 2.9, -2.9)$ of $Ax = b$ when

$$A = \begin{bmatrix} 4 & 1 & 1 \\ -1 & 3 & 1 \\ 0 & -1 & -2 \end{bmatrix} \qquad \text{and} \qquad b = \begin{bmatrix} -5 \\ 7 \\ 3 \end{bmatrix}.$$

Decomposing

$$A = S - T = \begin{bmatrix} 4 & & \\ & 3 & \\ & & -2 \end{bmatrix} - \begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

we get

$$S^{-1} = \begin{bmatrix} 1/4 & & \\ & 1/3 & \\ & & -1/2 \end{bmatrix} \text{ and } S^{-1}T = \begin{bmatrix} 0 & -1/4 & -1/4 \\ 1/3 & 0 & -1/3 \\ 0 & -1/2 & 0 \end{bmatrix}.$$

The norm $\|S^{-1}T\|$ is about .587, so our error at least (approximately) half at each iteration.

Start with the guess $x_1 = (1, 1, 1)$, so we don't know it, but $\|\varepsilon_1\| \approx 4.93$. We

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then calculate

$$\begin{aligned} x_2 &= S^{-1}Tx_1 + S^{-1}b \approx (-1.75, 2.33, -2) & \|\varepsilon_2\| &\approx 1.22 \\ x_3 &= S^{-1}Tx_2 + S^{-1}b \approx (-1.33, 2.41, -2.66) & \|\varepsilon_3\| &\approx .57 \\ x_4 &= S^{-1}Tx_3 + S^{-1}b \approx (-1.18, 2.77, -2.70) & \|\varepsilon_4\| &\approx .28 \end{aligned}$$

We will get error of less than .01 by about x_9.

The convergence here is a bit slow for a 3×3 matrix, but on a bigger matrix an approximation of %1 in 9 iterations is okay. This method works alright, but we need this $\|S^{-1}T\| < 1$, which can be a bit limiting. It is not to hard so see though that $\|S^{-1}T\| < 1$ if A is *strongly diagonal dominant* which means that for each row, (in absolute values) the diagonal entry is at least twice the some of all the entries in the row.

7.4.2 The Gauss-Seidel method

In the Gauss-Seidel method we let S be the lower triangular part of A.

Example 7.11. With the same A and b as in the previous example we get

$$A = S - T = \begin{bmatrix} 4 & & \\ -1 & 3 & \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 0 & -1 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$S^{-1} = \frac{1}{24}\begin{bmatrix} 6 & & \\ 2 & 8 & \\ -1 & -4 & -12 \end{bmatrix} \qquad S^{-1}T = \frac{-1}{24}\begin{bmatrix} 0 & 6 & 6 \\ 0 & 2 & 10 \\ 0 & -1 & -5 \end{bmatrix}$$

We compute $x_1 = (1, 1, 1)$, $x_2 = (-1.75, 1.42, 2.21)$, $x_3 = (-1.05, 2.72, -2.85)$. It is a little bit faster here, and indeed $\|S^{-1}T\| \approx .57$ is a little smaller than for the Jacobi decomposition. But for bigger matrices the improvement is usually can be much better. And we are much more likely to get a norm below 1.

Observe that in computing

$$x_2 = \frac{-1}{24}\begin{bmatrix} 0 & 6 & 6 \\ 0 & 2 & 10 \\ 0 & -1 & -5 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{24}\begin{bmatrix} 6 & & \\ 2 & 8 & \\ -1 & -4 & -12 \end{bmatrix}\begin{bmatrix} -5 \\ 7 \\ 3 \end{bmatrix}$$

the first coordinate is $\frac{-1}{24}(6+6)+\frac{1}{24}(-30) = -1.75$. Using this in place of the first co-oordinate 1 of x_1 gives an even better approximation of the second coordinate. Continuing in this way updating each coordinate of x_1 as we compute it, we speed up convergence significantly.

Problems from the text

7.4: 3, 6, 11, 14 (2, 8, 7, 12)

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8 Linear Programming and Game Theory

8.1 Linear Inequalities

A linear program (LP) consists of several constraints (linear equations or linear inequalities) in several variables, and a linear *objective* function in the variables. The goal is to maximise (or minimise) the objective function subject to the constraints.

Example 8.1. Maximize $x_1 + 2x_2 - x_3$ with respect to the constraints

$$\begin{array}{cccc} x_1 & +x_2 & +x_3 & \leq 10 \\ 2x_1 & -x_2 & & = 3 \\ & x_2 & +4x_3 & \geq 5 \\ x_1, x_2 & & & \geq 0 \end{array}$$

We start to explain the basic ideas with a simple LP, the following is a Maximum LP in Standard form as it can be written as "Maximize $c \cdot x$ subject to $Ax \leq b$ and $x \geq 0$" for an $m \times n$ matrix A, a vector x of variables, and vectors c and b of constants. (Here $Ax \leq b$ means that where a_i is the ith row of A, $a_i \cdot x \leq b_i$ for all i.)

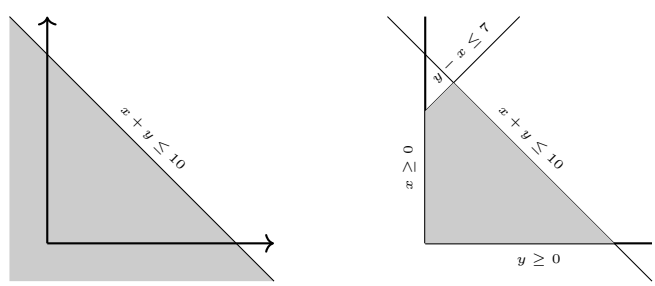
Practice

What are the A, b, c and x for the following (LP).

Example 8.2. A typical problem is to maximize $z = 2x + y$ with respect to the constraints

$$\begin{aligned} x + y &\leq 10 \\ y - x &\leq 7 \\ x, y &\geq 0 \end{aligned}$$

As a linear equation would restrict us to a line in $\mathbb{R}^2$, a linear inequality restricts us to a half space.



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The *feasible* points are the points that satisfy all constraints. The set of them is called the *feasible set*.

Practice

Draw the feasible points for the constraints $2x + y \geq 4$ and $2x + y \geq 3$. Draw them for $2x + y \geq 4$ and $2x - 2y \geq 4$.

Sometimes constraints can be redundant but generally each constraint 'cuts out a half plane'. The feasible set is always *convex*, which means for two points in the set, all points between them are also in the set. There are three distinct cases that can occur. The feasible set can be empty, bounded, or unbounded. In our previous example the feasible set was bounded. If we remove the constraint $x + y \leq 10$ then it would be unbounded.

Practice

Add a constraint that makes the feasible set empty.

If the feasible set is non-empty and bounded, then the objective function will have a maximum value on one (or more) of the corners. These corners are called *basic feasible solutions*. In the above example, we can simply evaluate $z = 2x + y$ at corners $(x, y) \in \{(0, 0), (0, 7), (10, 0), (1.5, 8.5)\}$ and find that its maximum is $z = 2(10) + (0) = 20$.

A more systematic way to do this is to consider the *isotopes* of $z = 2x + y$: the lines $2x + y = c$ for constants c. For the example above, the following are some of the isotopes:

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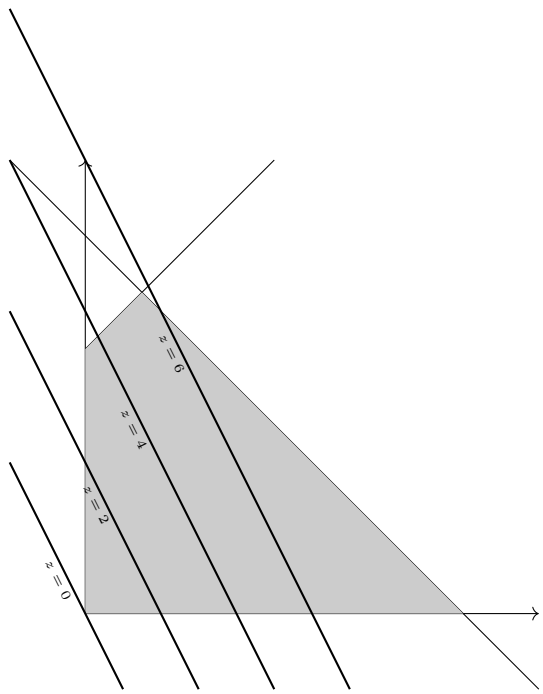

From this is is clear that on the feasible set, z is maximum at $(10, 0)$ and minimum at $(0, 0)$. We consider the vector orthogonal to the isotopes: the isotope $2x + y = 0$ is the line $y = -2x$ or $\langle(1, -2)\rangle$. The orthogonal vector is $(2, 1)$, and z increases in the positive direction. We choose the basic feasible solution 'farthest out' in that direction; ie. the corner whose projection onto $\langle(2, 1)\rangle$ is the greatest.

This seems simple enough, but an LP often has hundreds of variables and hundreds of contstraints. Finding all basic feasible solutions takes too much time. When there are n contraints and $m > n$ variables, each basic feasible solution occurs at the intersection of n constraints. There are m^n choices of the n hyperplanes to give such an intersection, only some of these are feasible. We want a better way than: trying all these intersection points, deciding which are feasible, and then evaluating z at them.

We want a systematic way to find the optimal basic feasible solution. There are several methods; the easiest is the simplex method. The idea for the simplex method is to start at basic feasible solution, and then look at 'nearby' basic feasible solution and move to the one maximises the increase in the objective function. The fortunate thing is that if we do this until there are no more local increases, then we have the optimum corner. The simplex method thus has two main Phases: in Phase 1 we find an initial feasible point (if one exists); in Phase 2 we find the optimal feasible point.

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8.1.1 Standard Form of an LP

We mentioned that an LP is in standard form is we can write it as "Maximize $c \cdot x$ wrt $Ax \leq b$ and $x \geq 0$." This means that we are to maximize some linear function $f(x) = c \cdot x$ with respect to two kinds of contraints:

- Positivity constraints $x_i \geq 0$ for each variable x_i, and
- general constraints of the form: $a_i \cdot x \leq b_i$.

The example we gave above was in standard form. For a problem in standard form, if the constants b_i are all positive, then the origin is a feasible point. The following example, which arises from a 'natural' problem, is not in standard form. But there are several simple methods to change constraints that are non-standard into standard constraints.

Maximise $f(x) = x_1 + x_2 + x_3$ with respect to the constraints

$$\begin{aligned} x_1 + 3x_2 + x_3 &\leq 2 \\ 2x_1 - x_2 + x_3 &\geq 4 \\ x_1 \geq 0, x_2 &\leq 0 \end{aligned}$$

We have a couple of problems

- The second constraint is $\geq$ rather than $\leq$.
- The constraint for x_2 is negativity rather than positivity.
- The variable x_3 doesn't have a positivity constraint.

First we deal with the negativity of x_2 by replacing x_2 with $-x_2$. The problem becomes

Maximise $f(x) = x_1 - x_2 + x_3$ with respect to the constraints

$$\begin{aligned} x_1 - 3x_2 + x_3 &\leq 2 \\ 2x_1 + x_2 + x_3 &\geq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Then we deal with the missing positivity constraint by 'splitting' x_3 into positive x_3 and negative x_4. The problem becomes

Maximise $f(x) = x_1 - x_2 + x_3 - x_4$ with respect to the constraints

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$$x_1 - 3x_2 + x_3 - x_4 \leq 2$$
$$2x_1 + x_2 + x_3 - x_4 \geq 4$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Finally, we flip the second constraint to:

$$-2x_1 - x_2 - x_3 + x_4 \leq -4.$$

Putting our problem into standard form, we have increased the number of variables, so the feasible space gets a little difficult to draw.

8.1.2 Matrix Form

Though the LP in standard form can be expressed with matrices as $Ax \leq b$. For the simplex method we will want it in a different form, known as *matrix form*, or *equational form*: maximise $c \cdot x$ wrt $Ax = b$, and $x \geq 0$. We change inequalities to equalites.

To replace the constraint

$$x_1 - 3x_2 + x_3 - x_4 \leq 2$$

we introduce a new variable $s_1 = 2 - (x_1 - 3x_2 + x_3 - x_4)$, called a *slack* variable. Observe that is must be positive for the inequality to be satisfied. So this constraint becomes

$$x_1 - 3x_2 + x_3 - x_4 + s_1 = 2 \text{ and } s_1 \geq 0.$$

Similarily replacing $-2x_1 - x_2 - x_3 + x_4 \leq 4$ with

$$-2x_1 - x_2 - x_3 + x_4 + s_2 = -4 \text{ and } s_2 \geq 0,$$

our problem is to maximize $(1, -1, 1, -1, 0, 0) \cdot (x_1, x_2, x_3, x_4, s_1, s_2)$ over all solutions of

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & 0 \\ -2 & -1 & -1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ s_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

that are positive in every variable.

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Practice

Make an LP-problem for the following situation, and put it into standard form and then matrix form. Solve it using goemetric intuition.

GM makes Buicks and Caddies. All cars that they make sell, and the profit per car is \$3 and \$5 respectively. It takes 2 and 3 minutes respectively to make the cars. How many of each should they make per hour to maximise their profit. (Perhaps it is clear to you here that they should only make Caddies. Lets add one more contraint.) Environmental laws say that average miles per gallon of the cars they make should be at least 18. Buicks and Caddies get respectively 17 and 20.

Problems from the text

8.1: 1 - 9 (1-9)

8.2 The Simplex Method

We will re-order our constraints so that the positivity constraints come first, and give them names. We solve the following LP-problem in standard form.

Example 8.3. Maximise $z = x_1 + x_2$ with respect to the constraints

$$c_1 : x_1 \geq 0 \qquad c_2 : x_2 \geq 0 \qquad c_3 : 2x_1 + x_2 \leq 10 \qquad c_4 : x_1 + x_2 \leq 7$$

To prepare for the simplex method, we put this in matrix form by adding a slack variable x_i for the non-positivity constraint c_i for $i = 3, 4$.

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 10 \\ x_1 + x_2 + x_4 &= 7 \end{aligned}$$

Setting $c = (1, 1, 0, 0)$ the problem now is to maximize $z = c^T x$ over all positive solutions to $Ax = b$:

$$\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 7 \end{bmatrix}.$$

When our original standard form problem has m *original* (non-positivity) constraints, and n *original* variables we add m slack variables for a total of $n+m$ variables. So A will by an $m \times (m + n)$ matrix. Each variable is a measure of

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how far we are from equalilty on a corresponding contraint, and so a basic feasible solution, which is the intersection of n of the constraints hyperplanes, has a 0 for the n corresponding variables. A set B of m variables is a *basis* for a solution if the only variables with non-zero values are in B. Variables in B are called *basic*.

Theorem 8.4. *Given an LP in standard form with n variables and m non-positivity constraints, consider the LP in matrix form with $n+m$ variables. Every basic feasible solution has a (not necessarily unique) basis B of m variables, and set B of m variables is a basis for at most one basic feasible solution*

Recall that to solve our LP via the simplex method, Phase 1 is to find a first basic feasible solution, and Phase 2 is to find the maximum basic feasible solution. When our problem is a maximization problem in standard form with positive b, then we have an easy first basic feasibles solution. One can take the set of slack variables as the basis.

So, in our example, we can skip Phase 1. We have a basic feasible solution with basis $B_{3,4} = \{x_3, x_4\}$. To find the values of x_3 and x_4 in this solution, we solve the system for x_3 and x_4.

$$\begin{aligned} x_3 &= 10 - 2x_1 - x_2 \\ x_4 &= 7 - x_1 - x_2 \end{aligned}$$

As $x_1 = x_2 = 0$, we can read off our solution as $x = (0, 0, 10, 7)$. Our objective function has the value

$$z = c \cdot x = (1, 1, 0, 0) \cdot (0, 0, 10, 7) = 0,$$

We lay out all our information for each basic feasible solution in a orderly form, called a tableau.

$$\begin{array}{ccccc} x_3 & = & 10 & -2x_1 & -x_2 \\ x_4 & = & 7 & -x_1 & +x_2 \\ \hline z & = & 0 & +x_1 & +x_2 \end{array}$$

This is the tableau for the basis $B_{3,4} = \{x_3, x_4\}$. The top $m = 2$ rows are our (non positivity) contraints, which have been solved for the basic variables x_3 and x_4. The final row is our objective function. The basic variables are to the left of the equations, to the right we have only zero variables and values. One thus easily reads of that $x_3 = 10, x_4 = 7$ and $z = 0$.

Now, we want to find a new feasible basic solution with greater objective value. Looking at the objective row, we choose one of the variables with positive co-efficient. If we increase it, letting it *enter* the basis, the objective function

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will increase. Choose x_1. When we increase its value, we have to maintain equality in the other rows. In the first row, the ratio of the constant 10 to the co-efficient -2 of x_1 is $-10/2 = -5$. This says that we can increase x_1 by at most 5. To compensate we would have to reduce x_3 to 0. In the second row, the ratio is -7. This is a weaker constraint, the strongest one is from the x_3 row, so x_3 will become 0 and exit the basis. (Notice that a positive co-efficient for x_1 in a row would not constrain its increase at all, it is the rows with negative co-efficients for the entering variable that we look at.) Our new basic feasible solution has basis $B_{1,4}$. It has the following tableau.

$$\begin{array}{ccccc} x_1 & = & 5 & -x_2/2 & -x_3/2 \\ x_4 & = & 2 & -x_2/2 & +x_3/2 \\ \hline z & = & 5 & +x_2/2 & -x_3/2 \end{array}$$

As x_1 is entering and x_3 is leaving, solve the x_3 contraint for x_1:

$$x_3 = 10 - 2x_1 - x_2 \qquad \Longleftrightarrow \qquad x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}x_3.$$

Then we want to replace x_1 in the other rows with its now value, to remove it from the right side of the table:

$$x_4 = 7 - x_1 + x_2 = 7 - (5 - \frac{1}{2}x_2 - \frac{1}{2}x_3) = 2 + \frac{1}{2}x_2 + \frac{1}{2}x_3.$$

We do the same for z.

Okay. Now, looking at the objective row, there are still non-negative co-efficients. We continue: x_2 enters, and x_4 exits. (Compute the x_2 row first.)

$$\begin{array}{ccccc} x_1 & = & 3 & & +x_4 \\ x_2 & = & 4 & +x_3 & -2x_4 \\ \hline z & = & 7 & +0x_3 & -x_4 \end{array}$$

Now, looking at the bottom row, there are no variables with positive co-efficients. So we cannot increase our value, and we stop. We have a maximum basic feasible solution $x = (3, 4, 0, 0)$ with value $z = 7$.

Practice

Do one more step of the simplex method letting x_3 enter the basis. Show that you get another maximum basic feasible solution.

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Practice

Draw the feasible set for the problem we did in this section. Identify the feasible solutions for each tableau, notice how we travel along the boundaries of the feasible set. Was there a faster route to the optimal solution? Why didn't we use it? When there are more than one positive values in the objective row of the tableau, we have a choice for the entering variable, we just took the one of smallest index. Suggest a 'better' heuristic for deciding which one to choose.

Actually the simplex method doesn't stop until all co-efficients in the objective row are negative. If there are only zere and negative co-efficients, we could be at what is called a degeneracy. In this case we have to run through all possible choices of the variables with 0 co-efficents as basic variables. Doing so, we might get to a tableau with positive co-efficents in the objective row, from which we can move our of the degeneracy. One has to do this in an orderly fashion to avoid cycling. This is very rare in 'natural' applications and so often ignored, but is a concern in using linear programs that come out of combinatorial constructions.

8.2.1 Minimisation Problems and Phase 1

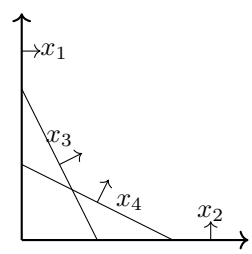
We have used the simplex method to solve a maximisation problem in standard form. As the problem had constraints $Ax \leq b$ where $b \geq 0$, we got an initial feasible basic solution by taking the slack variables as basic. When we do not have $b \geq 0$ this is not feasible. The following minimisation problem is such a problem.

Practice

Convert the following problem to standard form, and observe that the vector b is no longer positive.

Example 8.5. Minimise $x_1 + x_2$ subject to the constraints

$$\begin{aligned} x_1, x_2 &\geq 0 \\ x_1 + 2x_2 &\geq 6 \\ 2x_1 + x_2 &\geq 6 \end{aligned}$$



The feasible set is unbounded, but the objective function decreases towards that middle corner, so there will be a minimum, (but no maximum). To start the simplex method, we put our LP in matrix form.

Introducing slack variables x_3 and x_4 we get a problem (P) in matrix form.

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(P) Minimize $(1,1,0,0)^T x$ wrt $x \geq 0$ and

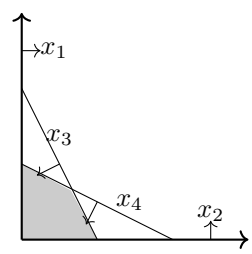
$$\begin{aligned} 2x_1 + x_2 - x_3 &= 6 \\ x_1 + 2x_2 - x_4 &= 6. \end{aligned}$$

Notice that the slack variables do not make-up a basic feasible solution: setting the other variables x_1 and x_2 to 0 we get that $x_3 = x_4 = -6$. This is not feasible. We have to get a feasible basic solution before we can start the simplex method.

8.2.2 Phase 1: Finding an initial basic feasible solution

We make an auxillary problem with a dual feasible set by 'flipping' the slack variables for the non-negativity constraints that are not satisfies by taking the slack variables as a basis. (In this case, it is all the slack variables.)

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 6 \\ 2x_1 + x_2 + x_4 &= 6 \\ x &\geq 0 \end{aligned}$$



This feasible set and the original one share a basic feasible solution with $x_3 = x_4 = 0$. To find this, it is enough to solve the following auxillary problem.

(A) Minimise $x_3 + x_4$ wrt

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 6 \\ 2x_1 + x_2 + x_4 &= 6 \\ x &\geq 0 \end{aligned}$$

For the problem (A) the slack variables do give a basic feasible solution. We solve it using the simplex method. The first tableau is:

$$\begin{array}{ccccc} x_3 & = & 6 & -x_1 & -2x_2 \\ x_4 & = & 6 & -2x_1 & -x_2 \\ \hline z & = & 12 & -3x_1 & -3x_2 \end{array}$$

As the problem we are solving is a 'Minimise' problem– we are trying to minimize the objective function– we are looking for negative co-efficients in the

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objective row. Choosing x_1 to enter we get that x_4 leaves, so the next tableau is:

$$\begin{array}{rcrrr} x_1 & = & 3 & -\frac{1}{2}x_2 & -\frac{1}{2}x_4 \\ x_3 & = & 3 & -\frac{3}{2}x_2 & +\frac{1}{2}x_4 \\ \hline z & = & 3 & -\frac{3}{2}x_2 & +\frac{3}{2}x_4 \end{array}$$

Now x_2 enters and x_3 leaves:

$$\begin{array}{rcrrr} x_1 & = & 2 & +\frac{1}{3}x_3 & -\frac{2}{3}x_4 \\ x_2 & = & 2 & -\frac{2}{3}x_3 & +\frac{1}{3}x_4 \\ \hline z & = & 0 & +x_3 & +x_4 \end{array}$$

Our minimal feasible solution is $x = (2, 2, 0, 0)$ with value 0. That the value is 0 means that the 'fake' variables introduces in the auxillary problem take a value of 0, so this is a solution to our original problem.

8.2.3 Phase 2

Phase 1 told us that $x = (2, 2, 0, 0)$ is a basic feasible solution to our problem (P). We start with a the basis $B_{1,2} = \{x_1, x_2\}$. Its tableau is

$$\begin{array}{rcrrr} x_1 & = & 2 & +\frac{2}{3}x_3 & -\frac{1}{3}x_4 \\ x_2 & = & 2 & -\frac{1}{3}x_3 & +\frac{2}{3}x_4 \\ \hline z & = & 4 & +\frac{1}{3}x_3 & +\frac{1}{3}x_4 \end{array}$$

We could compute this by solving for x_1 and x_2, but we already did this only we have replaced x_3 and x_4 with their opposites. Thus for the top of our tableau, we use the last tableau form Phase 1, and multiply the x_3 and x_4 columns by -1. The objective row we have to workout, as we are using a different objective function. $z = x_1 + x_2 = ...$ Hey look! Our objective row has only positive co-efficients. As we are trying to minimize z, we are done. That was lucky. This will not usually happen (well; it happens a lot in 2-dimensional examples.) In general, for Phase 2, you have to run the simplex method through again. Our miminum feasible solution $x = (2, 2, 0, 0)$, has value $z = 4$.

Let's finish off with one more example. This is the diet problem, we will use it in the next section.

Example 8.6. In a meal we need to eat 20g of protein and 10mg of iron, and we want to do so a cheaply as possible from a menu of eggs and beans. 100g of egg costs \$3 and provides 5g of protein and 2mg of iron. 100g of beans costs \$4 and provides 4g of protein and 3mg of iron. How much of each eggs and beans we eat?

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Letting x_1 and x_2 be the number of hundreds of grams of eggs and beans respectively, we want to minimize $(3,4)\cdot x$ over positive solutions to

$$\begin{aligned} 5x_1 + 4x_2 &\geq 20 \\ 2x_1 + 3x_2 &\geq 10 \end{aligned}$$

Adding slack variables x_3 and x_4, the set $\{x_3, x_4\}$ is not a basis to a basic feasible solution. So we use the auxillary problem to find an initial basic feasible solution:

(A) Minimise $x_3 + x_4$ over positive solutions to

$$\begin{aligned} 5x_1 + 4x_2 + x_3 &= 20 \\ 2x_1 + 3x_2 + x_4 &= 10. \end{aligned}$$

For this problem $B_{3,4}$ is the basis of a basic feasible solution. The tableau is

$$\begin{array}{ccccc} x_3 & = & 20 & -5x_1 & -4x_2 \\ x_4 & = & 10 & -2x_1 & -3x_2 \\ \hline z & = & 30 & -7x_1 & -7x_2 \end{array}$$

Choosing x_1 to enter the basis, x_3 has ratio -4 while x_4 has ratio -5, so x_3 leaves. The basis $B_{1,4}$ has tableau

$$\begin{array}{ccccc} x_1 & = & 4 & -\frac{4}{5}x_2 & -\frac{1}{5}x_3 \\ x_4 & = & 2 & -\frac{7}{5}x_2 & +\frac{2}{5}x_3 \\ \hline z & = & 2 & -\frac{7}{5}x_2 & +\frac{7}{5}x_3 \end{array}$$

Entering x_2 and leaving x_4 we get:

$$\begin{array}{ccccc} x_1 & = & \frac{20}{7} & -\frac{3}{7}x_3 & +\frac{4}{7}x_4 \\ x_2 & = & \frac{10}{7} & +\frac{2}{7}x_3 & -\frac{5}{7}x_4 \\ \hline z & = & 0 & +x_3 & +x_4 \end{array}$$

We have optimal solution $x = (\frac{20}{7}, \frac{30}{7}, 0, 0)$. This is our initial feasible solution for the main problem. Using the real objective function $(3,4)^T x$, the tableau for this is.

$$\begin{array}{ccccc} x_1 & = & \frac{20}{7} & +\frac{3}{7}x_3 & -\frac{4}{7}x_4 \\ x_2 & = & \frac{10}{7} & -\frac{2}{7}x_3 & +\frac{5}{7}x_4 \\ \hline z & = & \frac{100}{7} & +\frac{1}{7}x_3 & +\frac{8}{7}x_4 \end{array}$$

Again we are lucky, and find we have an optimal solution: $x = (20/7, 10/7, 0, 0)$, with cost $100/7$. We should each 285 grams of eggs and 142 grams of beans. Yum! This will cost us \$14.28.

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Problems from the text

8.2: 1, 2, 7, 8, 9.
Note that in #1, r is the cost row; $r > 0$ means that all entries are positive. (3, 8, 10, 1, 6)

8.3 The Dual Problem

For every (primal) LP-problem

$$(P) \text{ Minimise } c^T x \text{ subject to } Ax \geq b \text{ and } x \geq 0$$

there is a dual LP-problem

$$(D) \text{ Maximise } b^T y \text{ subject to } A^T y \leq c \text{ and } x \geq 0.$$

These two problems are closely related via the following theorem.

Theorem 8.7. *If (P) and (D) both have non-empty feasible sets then the minimum $c \cdot x$ over the feasible x for (P) is equal to the maximum $y \cdot b$ over the feasible y for (D).*

Before we prove this, lets see an example of a primal and dual problem.

Recall the example of the diet problem we had. We had to minimise $(3, 4) \cdot x$ over the positive solutions of

$$\begin{bmatrix} 5 & 4 \\ 2 & 3 \end{bmatrix} x \geq \begin{bmatrix} 20 \\ 10 \end{bmatrix}.$$

If this is our primal problem (P), then the following describes our dual problem (D).

There is a chemist who makes supplements of protein and iron. She must set the prices y_1 and y_2 of one unit (g for protein, mg for iron) of each of these supplements so as to maximise the amount she sells, with the understanding that people will only buy them if they are cheaper than buying beans and eggs. So, she wants to maximise $y \cdot (20, 10)$ over positive solutions of

$$5y_1 + 2y_2 \leq 3$$
$$4y_1 + 3y_2 \leq 4$$

The first constraint comes from the eggs. If the chemist doesn't satisfy this, it is cheaper to buy eggs. If she doesn't satisfy the second, it is cheaper to buy beans.

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Practice

Solve this problem with the simplex method. Show that the chemist should charge $(1/7, 8/7)$ for the respective amounts of (eggs, beans), to net \$14.24 per person. (Notice where this solution appears in the tableau of the primal problem.)

Before we prove Theorem 8.7, let's work out some equations for the tableau. Generally one does not use these when doing the simplex method, but one could.

Recall for a standard form LP (P') of minimising $c^T x$ wrt $Ax \geq b$ and $x \geq 0$, add slack variables to get the matrix form LP

(P) Minimize $\mathbf{c}^T\mathbf{x}$ wrt $\mathbf{A}\mathbf{x} = b$ and $\mathbf{x} \geq 0$.

For a basis $\mathcal{B}$ of a basic feasible solution, let B be the submatrix of $\mathbf{A}$ consisting of columns corresponding to indices in $\mathcal{B}$ and let N be the submatrix of columns not in N. So re-arranging columns, we can write $\mathcal{A} = [B \mid N]$ as the concatenation of B and N. Decompose $\mathbf{x}$ and $\mathbf{c}$ accordingly.

For example, if (P) were the problem of minimising $(0, 0, 1, 1)^T\mathbf{x}$ wrt

$$\begin{bmatrix} 2 & 1 & -1 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 7 \end{bmatrix},$$

then for basis $\mathcal{B} = \{3, 4\}$ we write

$$B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \; N = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \; \mathbf{x}_B = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} \; \mathbf{x}_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \; \mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \; \mathbf{c}_N = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Our constraint equation $\mathbf{A}\mathbf{x} = b$ can then be written as $B\mathbf{x}_B + N\mathbf{x}_N = b$.

Re-arranging this as $B\mathbf{x}_B = b - N\mathbf{x}_N$, and then solving for $\mathbf{x}_B$ we get

$$\mathbf{x}_B = B^{-1}b - B^{-1}N\mathbf{x}_N. \tag{9}$$

This is the top of our tableau. The first term here is the current value of $\mathbf{x}_B$ as all of the variables in $\mathbf{x}_N$ are 0. Similarily, we can write the objective function $z = \mathbf{c}^T\mathbf{x}$ as $z = \mathbf{c}_B^T\mathbf{x}_B + \mathbf{c}_N^T\mathbf{x}_N$. Inserting the value of $\mathbf{x}_B$ and then seperating 'value' terms and the 'zero' terms in $\mathbf{x}_N$ we get

$$z = \mathbf{c}_B^T B^{-1} b + (\mathbf{c}_N^T - \mathbf{c}_B^T B^{-1} N)\mathbf{x}_N. \tag{10}$$

This is the objective row of the tableau.

Practice

Verify this. For the example of the diet problem, choose a basic feasible solution, and compute the tableau using these formulas.

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Now we are ready for the proof.

Proof of Theorem 8.7. One part of the proof of this is easy. Indeed for any feasible points x of P and y of D we have

$$y^T b \leq y(Ax) = (yDA)x = (A^T y)^T x \leq c^T x.$$

The fact that there is a feasible point x for (P) means that the dual problem is bounded, and so has a maximum feasible solution. To finish the proof it is enough to find feasible points x and y such that $y^T b = c^T x$.

Adding slack variables (so $\mathbf{A} = [A \mid -I]$ because we are in standard form of a minimise problem) assume that $\mathbf{x}^*$ is the minimum basic feasible solution to (P), and $\mathcal{B}$ is its basis. We have by (9) that $\mathbf{x}^*_B = B^{-1}b$, $\mathbf{x}^*_N = 0$, and $z = \mathbf{c}^T\mathbf{x}^* = \mathbf{c}_B^T(B^{-1}b)$. Taking $y^* = (\mathbf{c}_B^T B^{-1})^T$ we have

$$(y^*)^T b = z = \mathbf{c}^T\mathbf{x}^* = c^T x^*$$

so it is enough to show that y^* is feasible.

To show that y^* is feasible we need to show that $A^T y^* \leq c$ and $y^* \geq 0$. We can write this as

$$\begin{bmatrix} A^T \\ -- \\ -I^T \end{bmatrix} y^* \leq \begin{bmatrix} c \\ -- \\ 0 \end{bmatrix}.$$

Re-arranging columns of $[A \mid -I] = \mathbf{A}$ so that $\mathbf{A} = [B \mid N]$ for the basis B of the solution $\mathbf{x}^*$ this becomes

$$\begin{bmatrix} B^T \\ -- \\ N^T \end{bmatrix} y^* \leq \begin{bmatrix} \mathbf{c}_B \\ -- \\ \mathbf{c}_N \end{bmatrix}.$$

Plugging in $y^* = (\mathbf{c}_b^T B^{-1})^T$ the top of this becomes $\mathbf{c}_B = (\mathbf{c}_B^T B^{-1} B)^T \leq \mathbf{c}_B$ which clearly holds, and the bottom becomes $(\mathbf{c}_B^T B^{-1} N)^T \leq \mathbf{c}_n$ which holds because by (10) the vector $\mathbf{c}_n^T - \mathbf{c}_B^T B^{-1} N$ is the vector of co-efficients in the objective row of the tableau for $\mathbf{c}^*$, and as $\mathbf{x}^*$ is a minimum solution, the co-efficients in the row must be positive.

□

Problems from the text

8.3: 2, 3, 5, 6, 7, 8, 9 (1, 3, 4, 11, 13, 12, 6)

8.4 Network Models

A network flow problem consists of a graph G with a set $V = V(G)$ of vertices (or nodes) and a set $E = E(G) \subset V \times V$ of edges, each edge $e \in E$ having a non-negative capacity c. For example:

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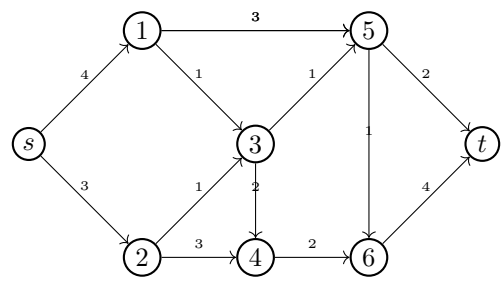

Here have vertex set $\{s, t, 1, 2, 3, 4, 5, 6\}$ and for the edge e_{ij} from i to j we have a capacity c_{ij}. So, for example $c_{15} = 3$. (If there is no edge from i to j it is useful to write $c_{ij} = 0$.) A *flow* is an assignment $f : E \to \mathbb{R}$ of 'flows' such that for each edge e_{ij} we have

$$0 \leq f(e_{ij}) \leq c_{ij},$$

and for each vertex $j \in V \setminus \{s, t\}$ flow through j is preserved:

$$\sum_i f(e_{ij}) = \sum_k f(e_{jk}).$$

The *value* of a flow f is the flow $|f| := \sum_i f(e_{si}) = \sum_j f(e_{jt})$ into t or out of s. (Using the preservation of flow at vertices, it is not hard to show that these must be the same.) The *max-flow* of the network flow problem is the maximum value $|f|$ over all flows f. We can find this with a linear program in several ways. The obvious way is to introduce a variable for each edge which denotes the flow along that edge. The flow from 1 to 5 is at most 3 so we would introduce a variable x_{15} and constrain it as

$$0 \leq x_{15} \leq 3.$$

The constraint at a vertex such as 1 would be $x_{s1} = x_{13} + x_{15}$.

In the pictured example, it is easy to find a flow of 3 or 4 or even 5. But is there a flow of 6? If not, how can we show it? Well, we use duality. First we introduce the dual problem.

Looking at the source s in the above example, it is clear that no flow f can have value $|f|$ more than 7, looking at the sink t it is clear that $|f| \leq 6$ for any flow f. This are easy bounds, but we can generalise them. A *cut* is a partition of the vertices into sets $S \ni s$ and $T \ni t$. The *value* of a cut (S, T) is

$$|(S, T)| := \sum_{i \in S, j \in T} c_{ij}.$$

The *min-cut* is minimum value $|(S, T)|$ over all cuts (S, T).

Practice

Can you find a cut of value 5 in the above network flow problem?

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It is clear that no flow f can have more flow across a cut (S, T) than the value of the cut: for $i \in S$ and $j \in T$ we have $f(e_{ij}) \leq c_{ij}$ so

$$|f| \leq \sum_{i \in S, j \in T} f(e_{ij}) \leq \sum_{i \in S, j \in T} c_{ij} = |(S, T)|.$$

That is, the value of any flow is at most the value of any cut, and in particular, the max-flow is at most the min-cut. The following theorem is called the Max-flow min-cut theorem. It is a special case of duality.

Theorem 8.8. *The max-flow of a network is equal to the min-cut.*

Now, to observe that the min-cut problem is the dual problem to the max-flow problem in the sense we defined in the last section, and so use the duality theorem to prove this, we must reformulate the max-cut problem by introducing variables for each path from s to t. Rather than this, prove the theorem with a graph theoretic argument.

Proof. Clearly max-flow is at most min-cut. We show equality by finding a flow f and a cut (S, T) with the same value. Let f be a flow of maximum value, and let S be the set of vertices v for which there is a path (called an augmenting path)

$$s = u_0 \to u_1 \to \ldots u_d = v$$

such that for each j

$$f(e_{u_i u_{i+1}}) < c_{u_i u_{i+1}} \text{ or } f(e_{u_{i+1} u_i}) > 0.$$

(The flow does not use all the capacity of the edge in the path, or it uses some but the edge is backwards in the path.)

Clearly $s \in S$ and $t \notin S$ or we could add more flow. Furthermore, the flow f must use all capacity of the cut $(S, V \setminus S)$; that is, for each $i \in S$ and $j \in T$,

$$f(e_{ij}) = c_{ij}.$$

If not, then we could extend the augmenting path from s to i into an augmenting path from s to j. But this contradicts our choice of S. Summing over all edges of the cut, we get that $|f| = |(S, V \setminus S)|$ as needed.

□

Problems from the text

8.4: 1, 4, 5 (4, 3, 2)

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8.5 Game Theory

Here is an example of a zero-sum game; 'zero-sum' referring to the fact that though money changes hand, the total money involved doesn't change.

Example 8.9. Players X and Y play rock-paper-scissors and at each round the loser pays the winner \$1.

What is player X's best strategy? How does he maximise the payout he gets? There is, of course, a lot of psychology involved in playing this game, but the point of view of game theory is that we are playing someone who is infinitely intelligent, and so our best chance against them is a random strategy.

A *(mixed) strategy* for the game is a vector (r, p, s) of probabilities such that the player will play rock with probability r, paper with probabiltiy p and scissors with probability s. We refer to a strategy for X as an X-strategy and one for Y as a Y-strategy.

Defining the payout matrix

$$A = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix},$$

for an X-strategy x and a Y-strategy y, player X expects to get a payout of $x^T Ay$ in every round and player Y expects a payout of $-x^T Ay$ each round.

Fixing a Y-strategy y, X can compute a *maximum expected payout* $\max_x x^T Ay$ for y, and a *best strategy* x_y *against* y that yields this expected payout.

Example 8.10. Y plays the strategy $y = (2/3, 1/3, 0)$ for rock-paper-scissors. We can use the following LP to find the best X-strategy x_y against y and the maximum expected payout $x_y^T Ay = \max_x x^T Ay$. Maximise $z = x^T(Ay) = (x_1, x_2, x_3) \cdot (-1/3, 2/3, -1/3)$ wrt $x_1 + x_2 + x_3 = 1$, $x_i \geq 0$. This is in equational form, and there is clearly a feasible solution taking any variable as basis. So we start with basis $\{x_1\}$, which has tableau:

$$\begin{array}{cccc} x_1 = & 1 & -x_2 & -x_3 \\ \hline z = & -\frac{1}{3} & +x_2 & +0x_3 \end{array}$$

x_2 enters and we end up with solution $x_y = (0, 1, 0)$ having value $2/3$.

Now, Y can also compute this maximum expected payout for every Y-strategy and so wants to play the Y-strategy y_* that minimises this. Playing y_* she can assure that X never gets payout greater than $\min_y \max_x x^T Ay$. Similarily X can compute the *mininum expected payout* $\min_y x^T Ay$ for every X-strategy and so plays strategy x_* that maximises this. We clearly have

$$\min_y (x_*^T Ay) \leq x_*^T Ay_* \leq \max_x x^T Ay_*$$

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Notice that for any x, taking $y = (1/3, 1/3, 1/3)$ gives a payout of 0, so $\min_y(\max_x x^T Ay) \leq 0$. While for any y, taking $x = (1/3, 1/3, 1/3)$ gives a payout of 0, so $0 \leq \max_x(\min_y x^T Ay)$. Thus

$$\max_x(\min_y x^T Ay) = 0 = \min_y(\max_x x^T Ay).$$

This is an *equilibrium*: $x_* = y_* = (1/3, 1/3, 1/3)$ for the game. The game has a *value* of 0. Neither player will stray from these strategies as doing so may (if the opponent notices and reacts) decrease their payout.

The minimax theorem says that this always happens. Any $m \times n$ matrix A defines a zero-sum game. An *X-strategy* for the game is a vector x of m probabililties $x_i \in (0, 1)$ that sum to 1, and a *Y-strategy* is a vector y of n probabilites that sum to 1. The *payout* for strategies x and y is the value $x^T Ay$. The *maximin* of the game is the X-strategy x_* such that

$$\min_y(x_*^T Ay) = \max_x \min_y(x^T Ay);$$

or equivalently, that for all X-strategies x,

$$\min_y(x^T Ay) \leq \min_y(x_*^T Ay).$$

The *minimax* is the Y-strategy y_* such that

$$\max_x x^T Ay_* = \min_x \max_y x^T Ay.$$

Theorem 8.11 (Minimax Theorem)**.** *For any matrix A,*

$$\min_y x_*^T Ay = x_*^T Ay_* = \max_x x^T Ay_*.$$

This value $x_*^T Ay_*$ is the value *of the game.*

Proof. Consider the following dual programs, where everywhere 1 is the vector of ones of appropriate length.

(P) Minimise $x^T 1$ wrt	(D) Maximise $1^T y$ wrt
$x^T A \geq 1^T$	$Ay \leq 1$
$x \geq 0$	$y \geq 0$

If these problems are both non-empty, then by duality we have solutions x' and y' such that $1^T x' = 1^T y' =: S$. The vectors $x_* := x'/S$ and $y_* = y'/S$ are vectors of probabilities each summing to 1, so are X- and Y-strategies respectively for the game. For any X-strategy x we have

$$\min_y(x^T Ay) \leq x^T Ay_* \leq \frac{x^T 1}{S} = 1/S$$

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while for any Y-strategy y we have

$$1/S = \frac{1^T y}{S} \leq x_*^T A y.$$

This latter holds for all y, so it certainly holds for the min y, and we get for any strategy x that

$$\min_y (x^T A y) \leq 1/S \leq \min_y (x_*^T A y),$$

showing that x_* is the maximin. Similarily we get that y_* is the minimax, and that $\max_x (x^T A y_*) \leq 1/S \leq \min_y (x_*^T A y)$. As we have in general for maximin x_* and minimax y_* that $\min_y (x_*^T A y) \leq x_*^T A y_* \leq \max_x (x^T A y_*)$, these values are equal and equal to $x_*^T A y_*$, as needed.

Now, the program (D) always has a feasible soultion $y = 0$. If (P) does not have a feasible solution, then for big enough a, we get a solution by adding a to each entry of A. This will not effect the strategies, as the payoff increases by the same amount for each outcome. This is enough.

□

A game is *fair* if its value is 0; if A is anti-symmetric: $A^T = -A$, then the game will be fair. Rock-paper-scissors is symmetric, so it is easy to see that its optimal strategy is a uniform vector of probabilities. This is not always true.

Example 8.12. Consider the game of rock-paper-scissors-521 where you win 5 points if you win with rock, 2 points if you win with paper, and 1 point if you win with scissors. The payout matrix for player X is

$$A = \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -1 \\ -5 & 1 & 0 \end{bmatrix}.$$

The game is still fair– neither player has an advantage– but $(1/3, 1/3, 1/3)$ is not the optimal strategy for either player? If X plays $x = (1/3, 1/3, 1/3)$ and Y plays $y = (1, 0, 0)$, then the expected payout is $\frac{1}{3}(2 - 5) = -1$.

Practice

Solve the linear program to show that X's optimal strategy against $y = (1, 0, 0)$ is $x = (0, 1, 0)$ with an expected payout of 2.

Using the dual programs from the proof of Theorem 8.11, we can find the equilibrium strategy for rock-paper-scissors-521.

Adding 2 to each entry of A makes $x = (1, 1, 1)$ a feasible solution to (P). We solve (D) with this adjusted A, so must maximise $y_1 + y_2 + y_3$ wrt

$$\begin{bmatrix} 2 & 0 & 7 \\ 4 & 2 & 1 \\ -3 & 3 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \leq \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

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The slack variables give us a feasible solution, so our first tableau is

$$\begin{array}{rcrrr} x_4 = & 1 & -2x_1 & & -7x_3 \\ x_5 = & 1 & -4x_1 & -2x_2 & -x_3 \\ x_6 = & 1 & +3x_1 & -3x_2 & -2x_3 \\ \hline z = & 0 & +x_1 & +x_2 & +x_3 \end{array}$$

The simplex method gives an optimal solution of $(1/8, 1/16, 1/4)$. This isn't a strategy but scaling it to be a strategy, we get that the optimal strategy is $x_* = y_* = (2/7, 1/7, 4/7)$. The value is $x_*^T A y_*$ using the unadjusted matrix A, but we don't have to compute this. The game is fail, so we know it will be 0.

Problems from the text

8.5: 4, 5, 7 (1, 6, 3)

References

[1] Gilbert Strang *Linear Algebra and its applications* Fourth Edition, International Student Edition. 2006 Thompson Learning.

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